CACTUS User Guide

Training Development

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1 Introduction

Common Authoring, Configuration, Transformation, Utilities and Services (CACTUS) is an XML-based authoring and publishing system developed by Commercial Aviation Services (CAS) Flight Services IT and used by multiple organizations to produce training material.

CACTUS is made up of two parts: a web application accessed through a browser, and an integration with an authoring tool, SyncRO Soft oXygen.

This user guide provides a detailed description of the Training Development application, from setup to assembly of the training material. For a more task-oriented guide, see the *CACTUS Training Development Quick Reference Guide*.

2 Overview

CACTUS supports content authoring in XML-based Darwin Information Typing Architecture (DITA). The primary components of DITA are topics and maps. Topics are the primary building blocks of content. Maps organize and combine topics and other content into chapters, which in turn may be combined into other groupings, such as volumes.

2.1 Data Stored in CACTUS

The following types of data are stored in CACTUS:

- **Topic files** are stand-alone information units that cover a single subject. The following topic types are supported:
  - Syllabus
  - Lesson Overview
  - Lesson Data
  - Lesson Plan
  - Assessment (Self)
  - Lesson Debrief
  - Procedures
  - Modules
  - Systems Knowledge
  - Glossary
- **Map files** combine topics into chapter units.
- **Book map files** combine chapters and front and back matter into books.
- **Images** include common graphic file formats: JPG, JPEG, PNG, AI, SVG, and FM.
- **Media files** include audio, video, Captivate simulations, and other media formats.

2.2 CACTUS Process

The following steps are used to create and publish training content in CACTUS:

1. Develop a training strategy and design.
2. Use the CACTUS workflow to initiate content development.
3. Use oXygen XML Author to create training content in DITA format.
4. Use the CACTUS publishing engine to assemble output.

2.3 Roles and Permissions in CACTUS

Permissions in CACTUS are controlled by the roles assigned to an account. Some roles are model-specific. A person can have multiple roles.

The following roles in CACTUS are model-specific:

- Lead
- Fleet Manager
- SME

The following roles in CACTUS are not model-specific:

- Editor
- Deployment
- Flight Standards (FS)
- Instructional Design (ISD)

Table 2-1 shows the permissions of each role.

Table 2-1 Permissions by Role

	Editor	Deployment	FS	FM	ISD	Lead	SME
Create/copy/ delete topic, map, book map, folder, upload media	X	X					
Perform oXygen functions (e.g., revision marking, applicability, insert objects/links)	X						
Check out, check in, cancel	X						
Acquire/unacquire workflow	X	X	X	X	X		X
Assign/unassign/ transfer workflow	X *					X **	
Mini-assemble	X	X	X	X	X	X	X
Draft/full assemble	X						

* For new content

** For edit, express, relabel

3 Getting Started

To get started with CACTUS, you will need the following:

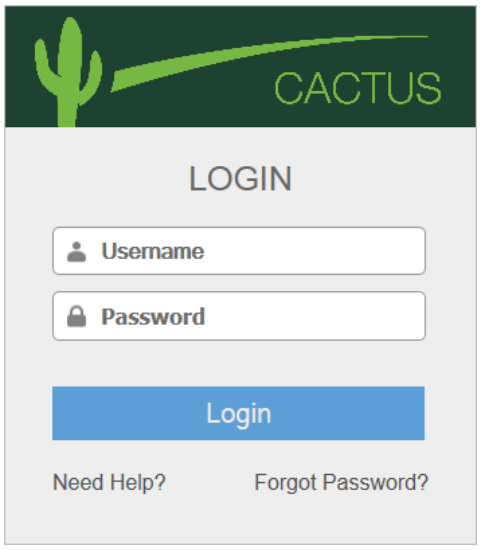
- CACTUS account – see the Requesting a CACTUS Account appendix
- oXygen XML Author software – see the Installing and Integrating oXygen appendix.
- Firefox web browser (recommended browser for CACTUS applications) – see the Installing Firefox and Java appendix.
- Java version 8 – see the Installing Firefox and Java appendix.

3.1 Logging in to CACTUS

1. Copy and paste this URL into a Firefox browser: https://cactus-app-prod.ca.boeing.com:9443/cactus/.

   For most users, the login will automatically be handled by WSSO. The user will click the CACTUS link and be immediately logged in. For others, such as the Systems Service Enterprises (SSE) contractors in the JAX media group, WSSO will not work.

   If a CACTUS login is required, the LOGIN dialog box will be displayed.



2. Enter your **Username** and **Password**, and then click **Login**.

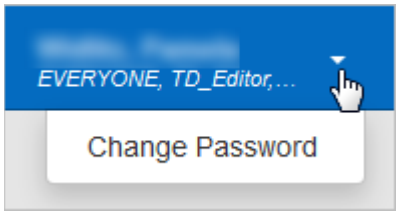
   The landing page will vary, depending on your role and organization.

3.1.1 Accessing Login Support

- Click **Need Help?** to open a login troubleshooting page.
- Click **Forgot Password?** to e-mail the CACTUS support team for assistance.

3.1.2 Resetting your Password

1. In the CACTUS menu bar, click the down arrow to the right of your user name and role, and then click **Change Password**.



   The **Change Password** dialog box is displayed.

2. Enter your old and new passwords, re-enter your new password to confirm, and then click **Save**.

4 Touring the CACTUS Interface

The CACTUS web application interface is shown in Figure 4-1.

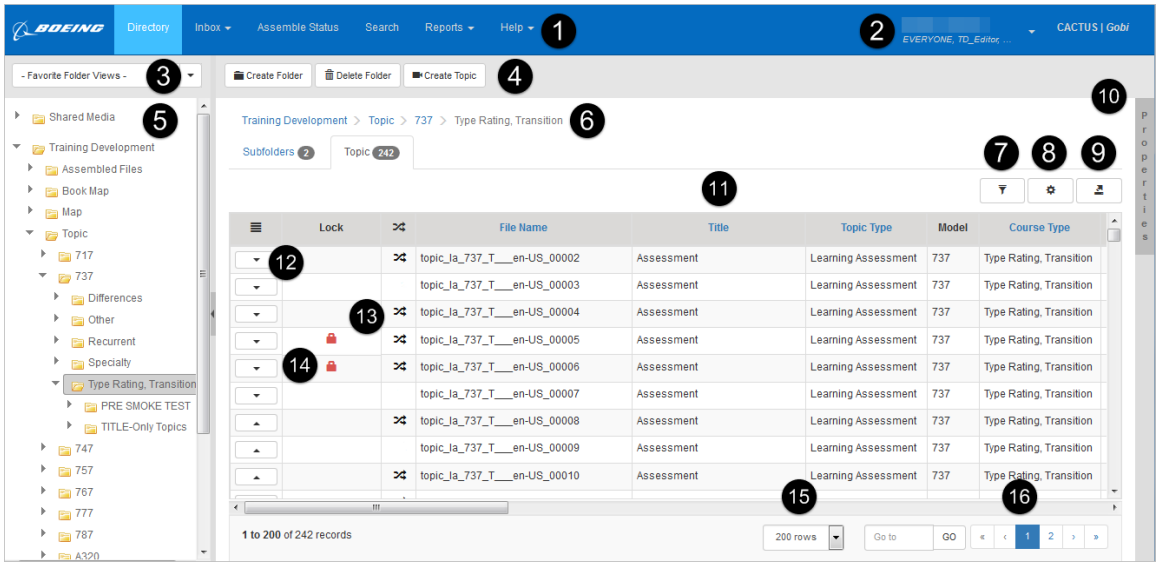


Figure 4-1 CACTUS Interface

Key

1.	Menu bar	Provides access to the main functions of the CACTUS application.
2.	Username/role	Displays the user's login information.
3.	Views	Allows you to select views that you have created and saved.
4.	Create/delete buttons	Allows users to create and delete folders and create topics.
5.	Directory pane	Allows you to navigate the directory tree to find folders and files.
6.	Navigation breadcrumbs	Allows you to see your path location in the directory and to navigate to folders in the path.
7.	Grid filter	Allows you to create and apply filters to the grid display.
8.	Grid preferences	Allows you to customize your grid display (e.g., selecting and ordering columns).
9.	Export to Excel	Allows you to export the current grid to Excel.
10.	Properties pane (closed position)	When opened, provides information about the selected file or folder (e.g., Created By, Modified Date); to display the Properties pane, select a row in the grid, and then click **Properties** on the right edge of the window.
11.	Grid	Displays the contents of the folder that is selected in the directory tree.
12.	Action menu	Provides commands to operate on the file in the selected row (e.g., Check Out, Copy)
13.	Workflow indicator	Identifies an object that has a workflow associated to it; clicking the workflow icon opens a workflow dialog box, which provides information (e.g. history, status) of the object
14.	Lock indicator	Identifies an object that is checked out
15.	Number of rows displayed	Allows you to set the number of rows displayed
16.	Page-to-page navigation	Allows you to navigate to specific pages

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5 Using the Directory Pane

This segment describes how to use the directory tree, as well as how to save, display, update, and delete directory views.

5.1 Navigating the Directory Tree

Clicking the triangle beside a folder will expand the folder in the tree.

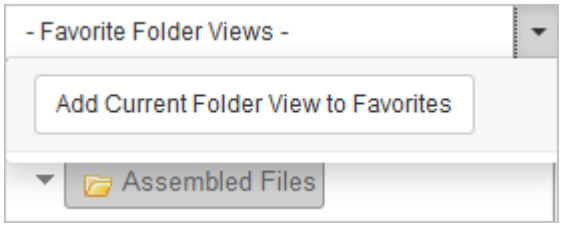


Clicking a folder will expand the folder and display the contents of the folder in the main grid.

5.2 Saving a Directory View

A saved directory view stores the display of folders that have been expanded in the directory tree. Saving a directory view allows you to quickly navigate to specific folders. The saved view that you used last will automatically be loaded when you log in to CACTUS. The currently loaded view will be displayed in the field above the tree.

1. Navigate to a folder in the tree.
2. Click the down arrow in the field above the tree:



3. Select **Add Current Folder View to Favorites**.
4. Type a name for the view in the **Add New Favorite View** dialog box.

5.3 Displaying a Saved View

1. Click the view down arrow in the field above the tree.
2. In the list, select the view to load.

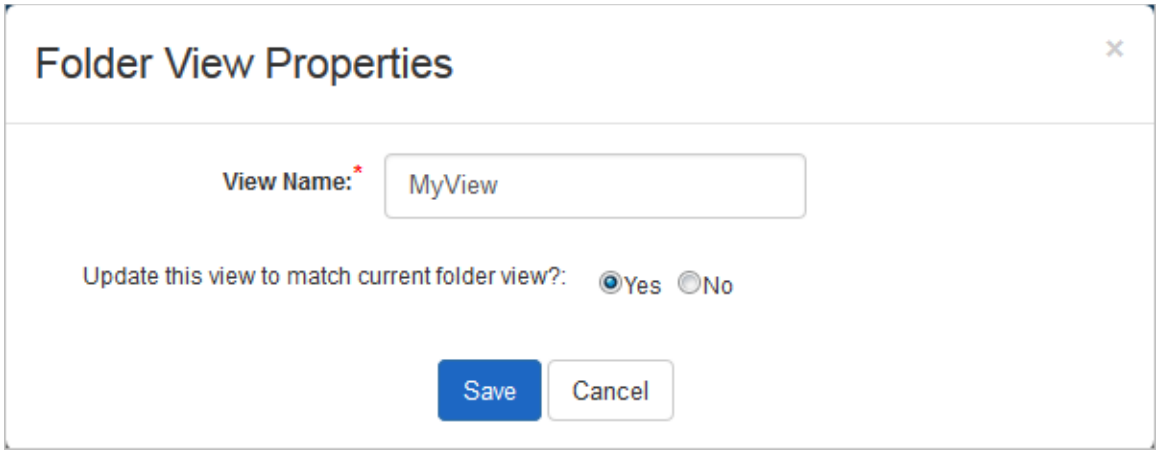
   Note: You must have two views saved to load an existing view.

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5.4 Updating a Directory View

1. Expand/collapse the folders in the tree to create the view you want to save.
2. Click the down arrow in the field above the tree.
3. In the list, select the pencil icon for the view that you want to update.

   The **Folder View Properties** dialog box is displayed.



4. Keep/update the name of the view.
5. Select whether you want to update the view to match the current folder view.
6. Click **Save**.

5.5 Deleting a Directory View

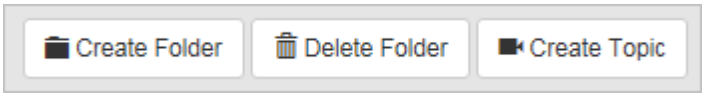
1. Click the view down arrow in the field above the tree.
2. Select the trash can icon for the view you want to delete.
3. In the confirmation dialog box, click **Yes**.

6 Using the Grid

This segment describes how to use the components of the main CACTUS grid.

6.1 Submenu Buttons

The buttons above the grid and below the menu bar allow you to create and delete folders and files. You must navigate down to lower level folders for the Create/Delete buttons to by displayed. The Create buttons display dialog boxes to allow you to enter required and optional metadata.



Note: See the

Special Characters appendix for a list of characters not allowed in property fields, such as title, keywords, and comments.

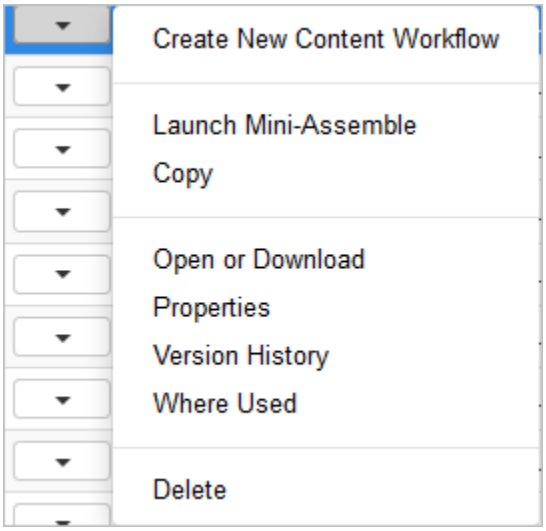
6.2 Navigation Breadcrumbs

The navigation breadcrumbs above the grid display the path to the current folder in the directory tree. By clicking the hyperlinks in the breadcrumbs, you can quickly navigate to folders in the path.



6.3 Action Menu

The Action menu is displayed by clicking the down arrow at the left end of a row. It provides operations you can perform on the selected file. Menu options are determined by your role and status of the object. The screenshot below shows an Action menu that an Editor might see for a topic that is in the SME Review state.



6.4 Changing Pages

You can move sequentially through pages in the grid or jump to a specific page.

6.4.1 Moving Between Pages

Below the grid, select the page number to display. Use the **<<** and **>>** buttons to move to the first or last page.

6.4.2 Going to a Specific Page

1. In the **Go to** box, type the number of the page to display.
2. Click **GO** or press Enter.

6.5 Resizing Columns

To resize columns, drag the border between columns to the left or right.

6.6 Sorting Columns

Click any blue column header to sort the contents of the grid by that column. Clicking it again reverses the sort order.

Note: You cannot sort by clicking black column headers.

6.7 Filtering on Columns

A filter shows/hides items in the grid, based on filter criteria.

Note: Only columns with headings in blue text can be filtered.

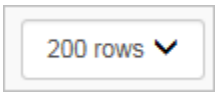
1. Navigate to a folder in the tree.
2. Click the **Toggle Grid Filters** button (left button above the grid):



3. Enter a value in the text field above a column of the grid.
4. Click **Apply Filter** or press Enter.
    - To remove the values from all filter boxes, click **Clear Filters**.
    - To hide the filter boxes, click the **Toggle Grid Filters** button.

6.8 Changing the Number of Rows Displayed

Below the grid, change the left drop-down list to the number of rows to be displayed:



Note: All files may not display in one screen. If you do not see a file in the initial grid display, it may be listed in a subsequent screen. Use the navigation features described in Moving Between Pages and Going to a Specific Page to move through the remaining screens.

6.9 Setting Grid Preferences

On the grid preference screen, you can change the column order, hide columns, and set default filter values.

1. Navigate to a folder in the tree.
2. Click the **Grid Preference** button (gear button above grid):



6.9.1 Changing the Column Order

1. In the **Grid Preferences** dialog box, select the row for the column that you want to move.

2. Use the **Top**, **Up**, **Down**, and **Bottom** buttons to move the row.

6.9.2 Showing/Hiding Columns

In the **Grid Preferences** dialog box, select/clear the **Display** checkbox for the column row you want to show/hide.

6.9.3 Freezing/Unfreezing Columns

Freezing a column keeps it visible when you scroll horizontally.

1. To freeze a column, in the **Grid Preferences** dialog box, select the radio button in the **Freeze** column that corresponds to the column you want to remain visible.

2. Click **Apply**.

Note: When you freeze a column, any columns to the left of that column are also frozen.

To unfreeze a column, click **Unfreeze**, and then **Apply**.

6.9.4 Setting Default Filter Values

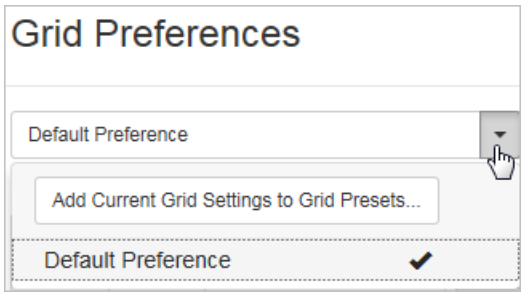
Enter a value in the default filter box for the row.

When a default filter is set, the grid will automatically apply the filter to that column when a grid with the data type is displayed.

6.10 Saving Grid Preferences

1. Set your preferences in the **Grid Preferences** dialog box.

2. At the top of the **Grid Preferences** dialog box, click the down arrow to the right of the view list box.



3. Click **Add Current Grid Settings to Grid Presets**.

4. Enter a name for the view.
5. Click **Save**.

6.11 Loading Grid Preferences

At the top of the **Grid Preferences** dialog box, select a view from the drop-down list.

6.12 Updating Grid Preferences

1. Load the view you want to update.
2. Update your grid preferences.
3. Click the down arrow to the right of the view list at the top of the dialog box.
4. You can change the grid preference one of two ways:
   - Change the name of the view if desired by clicking **Add Current Grid Settings to Grid Presets**.

     or

   - Click the pencil button. You can change the grid preset name or leave it unchanged. Leave **Yes** selected**.**
5. Click **Save**.
6. Click **Yes** to confirm.

6.13 Deleting Grid Preferences

1. On the grid, click the Grid Preferences icon on the right side of the screen, above the grid.



2. In the **Grid Preferences** dialog box, click the down arrow, and then click the trash icon next to the custom grid.



3. Click **Yes** to confirm.

6.14 Exporting the Grid to Excel

1. Navigate to a folder in the tree.
2. Click the **Export to Excel** button (right button above the grid):

   The current grid view is exported to an Excel spreadsheet.

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7 Searching

You can search on the properties of files to find only those files that match search criteria. In the search results, depending on your role, you can perform certain standard grid actions, such as sorting and setting preferences, by using the Action menu in the first column.

1. Click **Search** on the menu bar.
2. In the **Search Criteria** screen, enter your search parameters.
3. Click **Search**.

To reset all fields to their default values, click the **Clear** button.

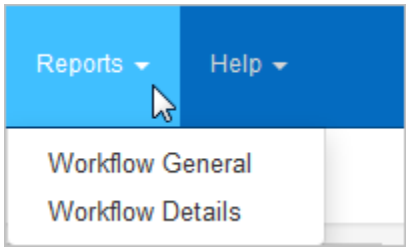
Once a search is performed, the search criteria are hidden and the search results are displayed. Click **Hide Criteria** or **Show Criteria** to toggle the display of the search criteria.

You can export the results of a search to your computer as an Excel file. To export search results, click the **Export to Excel** button, which is above the search grid:



8 Viewing Reports

1. On the CACTUS menu bar, click **Reports**, and then choose a report type.

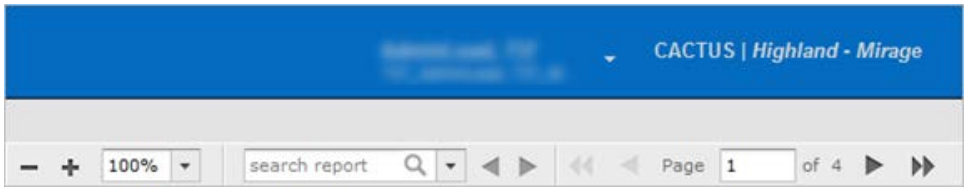


   An **Options** screen for the selected report is displayed.

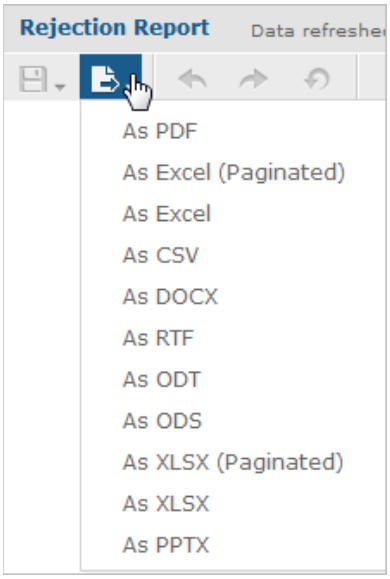
2. Enter the required information and optional information as needed, and then click **Apply** in the lower left corner of the screen.

   The report is displayed.

   - Use the fields at the top of the screen to adjust the display, search, and navigate to other pages in the report.



   - To export the report, click the export icon, and then select a format from the drop-down list.



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9 Managing Files and Folders

This segment describes how to create and perform operations on folders and files. These operations are performed on the folder and file object level.

9.1 Creating a Folder

CACTUS allows new folders to be created only below a specific level of the directory tree. Folders can be created only under a Course Type or Media Type level folder (two levels under the model folder or lower).

The **Create Folder** button will be displayed only if you are in a location where a folder can be created and you have write permission to the location.

1. Navigate to the folder where you can create a new subfolder.
2. Click the **Create Folder** button below the menu bar.
3. Type the name of the folder and a title for the folder.
4. Click **Finish.**

9.2 Deleting a Folder

Only user-created, empty directory folders can be deleted. Default system-created folders (e.g., model folders, media type folders) cannot be deleted.

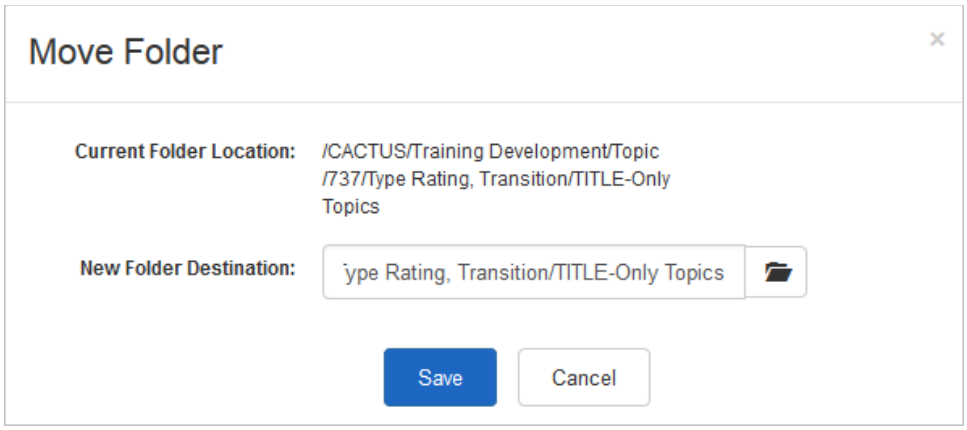
1. Navigate down through the folders until you are inside the folder you want to delete.

   The folder must be empty before you will be allowed to delete it.
2. Click the **Delete Folder** button below the menu bar.
3. Click **Yes** to confirm.

9.3 Moving a Folder

1. Navigate to the folder you want to move.
2. Click the Action menu for that folder, and choose **Move Folder**.

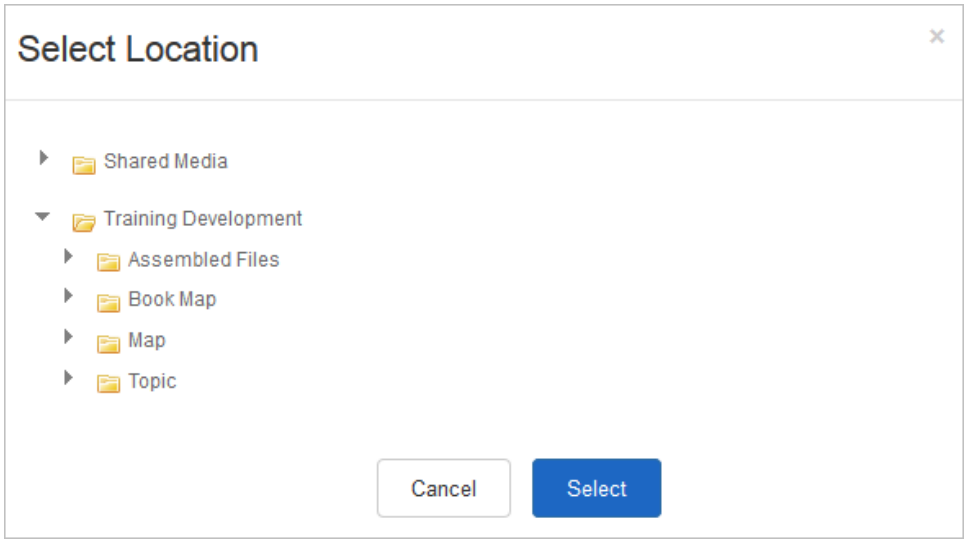
   The **Move Folder** dialog box is displayed.



3. Click the folder icon to select a new location.

   Note: The location defaults to the current folder location.

   The **Select Location** dialog box is displayed.



4. Navigate to the new folder location, and then click **Select**.

   The **Move Folder** dialog box is displayed with the selected location in the **New Folder Destination** field.

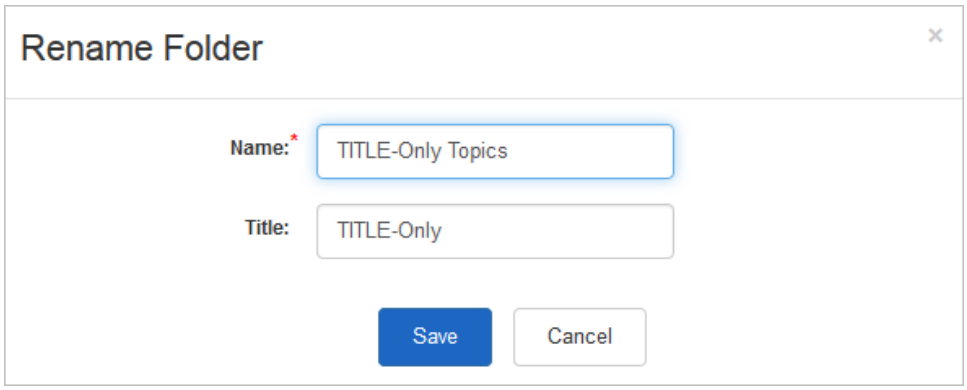
5. Click **Save**.

9.4 Renaming a Folder

1. Navigate to the folder you want to rename.
2. Click the Action menu for that folder, and choose **Rename**.

   The **Rename Folder** dialog box is displayed.

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3. Enter a new name and optional title, and then click **Save**.

   The grid refreshes and displays the folder with its updated name.

9.5 Creating a File

Files can be created only in folders or subfolders that are appropriate to their type. For example, book maps can be created only in Book Map folders, topics in Topic folders, and so on.

To create a book map, map, or topic file (TD Editor role):

1. Navigate to the folder where the file will be saved.
2. Click the appropriate **Create** button below the menu bar.
3. Enter the required properties.

Note: See the

Special Characters appendix for a list of characters not allowed in property fields, such as title, keywords, and comments.

4. Click **Create.**

Upload a media file (TD Editor role):

1. Navigate to a media folder where the media file will be uploaded.

   Note: Media can be uploaded only to media type folders or their subfolders.

2. Click **Upload Media** below the menu bar.

3. In the **Upload Media** dialog box, navigate to the file to be uploaded.

4. Enter the media file properties.

Note: See the

Special Characters appendix for a list of characters not allowed in property fields, such as title, keywords, and comments.

5. Click **Upload**.

9.6 Saving a File

When you create a new file, upload a media file, or check in a file that you have checked out from a workflow, you are saving the file to the repository.

9.7 Copying a File

You can use the Copy command to create a new file from an existing file. When you perform a copy, the properties of the existing file are shown. You can edit the properties to create the properties of the new file. The new file will have the same contents as the existing file. After the file is copied, a success message is displayed. The message contains a link to the folder where the new file was saved.

Copy a book map, map, or topic file:

1. Navigate to the folder that contains the file you want to copy (TD Editor role).
2. On the Action menu, choose **Copy**.
3. Skip this step if you want to save the new file in the same location as the original file. The destination folder defaults to the same folder as the existing file.

   Change **Destination Folder** to the location where the new file will be saved.

   Navigate to the folder that contains the file you want to copy (TD Editor role).
4. Change the values of the properties as desired.
5. Click **Create.**

9.8 Renaming a File

You can only rename a file when you check it in. See Checking In a File.

9.9 Viewing a File

1. Navigate to the folder that contains the file you want to view.
2. On the **Action** menu, choose **Open** or **Download**.

   A dialog box is displayed, providing an option to save the file or open it.

9.10 Checking Out a File

You can check out a file by using the Action menu in a workflow grid if the file is associated to a workflow.

If the file is associated to a workflow, you can check out the file from the workflow details screen by using the Action menu. You can open a workflow details screen and workflow grid by locating the workflow in an inbox, or by clicking on the workflow icon in the main grid.

When a file is checked out, a new version is created with the value of *pwc* (personal working copy) in the Version column in the main grid. This is a temporary copy created by the repository. The grid shows both the latest version and the pwc version.

Note: When files are checked out, they may take a while to open.

From a workflow details screen:

1. On the menu bar, click **Inbox**.
2. Navigate to the workflow inbox that contains the file that you want to check out.
3. Open the workflow and select the file you want to check out from the workflow grid.
4. On the Action menu, choose **Check Out**.

   The file is checked out and downloaded to your computer.

From the main grid:

1. Navigate to the file you want to check out.
2. Click on the workflow icon next to the file.

   The workflow details screen and workflow grid are displayed.
3. On the Action menu, choose **Check Out**.

   The file is checked out and downloaded it to your computer.

9.11 Editing a File

This function is available to the TD Editor role.

1. In CACTUS, check out the file by choosing **Check Out** on the Action menu in the workflow dialog box.

   A pwc version of the file is downloaded to the folder C:\Users\<*userid*>\Documents\cactus.

   On your computer, the file is opened in oXygen.

2. Update the file, then save the file, and close the file in oXygen.

3. In CACTUS, check in the file. You can update the properties during check-in.

   The file on your computer is saved to the CACTUS repository as a new version.

9.12 Checking In a File

The Check In command is available only if you have files checked out. You can check in a file by using the Action menu in a workflow grid if the file is associated to a workflow.

If the file is associated to a workflow, you can check in the file from the workflow details screen by using the Action menu. You can open a workflow details screen and workflow grid by locating the workflow in an inbox, or by clicking on the workflow icon in the main grid.

Note: When you check in a file, the pwc version is removed from your computer.

From a workflow details screen:

1. On the menu bar, click **Inbox**.

2. Navigate to the folder that contains the file that you want to check in.

3. Open the workflow and select the file you want to check in from the workflow grid.

4. On the Action menu of the file, choose **Check In**.

5. Update the properties as desired.

6. Click **Check In**.

From the main grid:

1. Navigate to the file.

2. In the Action menu of the pwc version of the file, choose **Check In**.

3. Update the properties as necessary.

4. Click **Check In**.

9.13 Canceling a Checkout

The Cancel Checkout option is available only if you have files checked out. Canceling a checkout removes the checkout from the file and deletes the downloaded file.

Caution: When you cancel a checkout, any changes made to the file will be lost and cannot be recovered.

You can cancel a file checkout either from a workflow details screen or from the main grid by using the Action menu. You can open a workflow details screen and workflow grid by locating the workflow in an inbox, or by clicking on the workflow icon in the main grid.

Note: When you cancel a checkout, the file is removed from your computer.

From a workflow details screen:

1. On the menu bar, click **Inbox**.

2. Navigate to the workflow inbox that contains the file you want to check in.

3. Open the workflow and select the file for which you want to cancel the checkout from the workflow grid.

4. On the Action menu, choose **Cancel Checkout**.

5. Click **Yes** in the confirmation dialog box.

From the main grid:

1. Navigate to the checked out file.

2. Click on the workflow icon  next to the file.

   The workflow details screen and workflow grid are displayed.

3. On the Action menu of the pwc version of the file, choose **Cancel Checkout.**

4. Click **Yes** in the confirmation dialog box.

9.14 Deleting a File

Only the Editor and Deployment roles can delete a file. Files that are referenced in a map or book map, are associated to an active workflow, or have had Approved status assigned cannot be deleted.

Caution: When a file is deleted, it cannot be restored.

1. Navigate to the file to be deleted.
2. On the Action menu, choose **Delete**.
3. Click **Yes** in the confirmation dialog box.

10 Performing Assemblies

This segment describes how to preview and publish content using the assemble functionality. There are three types of assemblies:

- Mini-assemble
- Draft assemble
- Full assemble

After you perform an assemble, you can check its status by opening the Assemble Status screen. See Viewing Assemble Status.

10.1 Performing a Mini-Assemble

Mini-assemble is performed on one topic/chapter map file and is comparable to a "print preview." It is removed from the **Assemble Status** screen after 30 days.

1. Navigate to the file to be assembled.
2. On the Action menu, choose **Launch Mini-Assemble**.
3. Enter the required assemble information.
4. Click **Launch**.

10.2 Performing a Draft Assemble

A draft assemble collects the topics, maps and media into a single publication. It is removed from the **Assemble Status** screen after 30 days.

1. Navigate to the workflow inbox and open the workflow to assemble.
2. On the workflow details screen, click **Launch Draft Assemble**.
3. In the Draft **Assemble** dialog box, select the Type of Manual and whether revision marking should be included, and then click **Start Draft Assemble**.

10.3 Performing a Full Assemble

A full assemble is similar to a draft assemble, except that it is stored indefinitely in the Assembled Files folder in the CACTUS repository.

1. Navigate to the workflow inbox and open the workflow to assemble.
2. On the workflow details screen, in the **Workflow Activity** area, enter **Time Tracking** before you launch an assemble.

3. On the workflow details screen, click **Launch Assemble**.

4. In the **Launch Assemble** dialog box, select the Delivery Type, Type of Manual, and whether revision marking should be included, and then click **Assemble**.

5. Reopen the workflow; note that the workflow is now in the Editor Review step. Click the **Assembled File** tab to view the assembled output.

6. On the Action menu, you can select to **Open or Download** the file, or to **View the Log**.

7. You can also view the file from the **Assembled Status** tab. See Viewing Assemble Status.

10.4 Viewing Assemble Status

The Assemble Status screen shows a list of all assembles (including assembles, draft assembles, and mini-assembles) that have been performed in the last 30 days. The screen shows whether an assemble completed successfully and allows you to view the error log.

Note: You may need to refresh the status screen by clicking the **Grid Refresh** icon located at the far right of the screen to see the status update.

1. Click the **Assemble Status** tab.

   The **Assemble Status** screen is displayed.

2. View the **Status** column to see whether the assembly succeeded, failed, or is still in progress.

3. To view the error log, click the **View Log** link for the mini-assemble.

4. To download mini-assemble files, click the download link (the blue down arrow in the first column).

11 Using Workflows: Training Development

A workflow is created to track a related set of work. No content work is allowed without a workflow.

11.1 Setting Preferences for the Workflow Grid

To display the workflow details screen and workflow grid, click **Inbox** on the menu bar.

You can set display preferences for the workflow grid, including column order and default filter values. See Setting Grid Preferences.

11.2 Filtering the Workflow Grid

To display the workflow details screen and workflow grid, click **Inbox** on the menu bar.

You can limit the items displayed in the workflow grid only to those that match a specific value in a column. See Filtering on Columns.

11.3 Starting a New Workflow

The table below describes the four types of workflows:

Workflow Type	Creator	Application
New Content	TD Editor	For creating a new, large manual in smaller pieces (maps/chapters). Allows several editors, SMEs, and ISDs to work on the same manual at the same time, just in different chapters.
Editing	TD Lead	For creating small manuals, revising and existing manual, or making a new version/copy of an existing manual.
Express	TD Lead	Shortened Editing workflow for creating a minor revision, already approved by Fleet Manager and Flight Standards, to an existing manual. Helps to expedite corrections or urgent revisions.
Relabel	TD Lead	For a customer who wants to take an existing Boeing Standard Course and re-label it with their name (i.e., "The Boeing Company" is replaced with the customer name, and “TBC” is replaced with the customer’s code). Eliminates check-out or versioning of the XML files, as content is not changed.

1. On the menu bar, click **Directory** to display the folder grid.
2. Navigate the directory tree to locate the book map (or Map/Topic for Editors) for which you want to start a workflow .
3. On the Action menu of the row where you want to create the workflow, choose **Create (New Content) Workflow**.

4. For TD Leads, select the type of workflow, and then click **Continue**.

5. In the Create Workflow dialog box, enter the relevant information, and then click **Create**.

11.4 Finding a Workflow

There are two ways to find and open workflows:

- In the inboxes.
- From an icon in the main grid.

11.4.1 Finding a Workflow in the Inboxes

The first way to find a workflow is by looking in one of the four types of inboxes. You can access all of these by clicking **Inbox** on the menu bar.

- Find a workflow assigned to you:

  Click the **My Inbox** tab.

- Find a workflow that was assigned to you at any time, including currently:

  Click the **Involved Workflows** tab.

  To see workflows that have been closed in the last 30 days, select **Closed Workflows**.

- Find a workflow assigned to your group:

  Click the tab matching your workgroup's queue.

  To narrow down the list of workflows, you can search by **Model** and **Assigned To**.

- Find any workflow:

  Click the **All Workflows** tab.

  To narrow down the list of workflows, you can search by **Model**, **Assigned To**, and **Workflow Status**.

11.4.2 Finding a Workflow in the Main Grid

If a file is in a workflow, a workflow icon will be displayed in the row in the main grid. Clicking the icon will open the workflow details screen and the workflow grid.

11.5 Opening a Workflow

When you open a workflow, you will have edit permission if the workflow is currently assigned to you. You will have read-only permission if the workflow is not assigned to you.

You have two ways to open a workflow:

- From your inbox, choose **Open** from the Action menu, or click the link in the **Workflow Name** column.
- In the main grid, click the workflow icon.

11.6 Acquiring a Workflow

Acquiring a workflow assigns it to you from the workgroup's queue. After you acquire a workflow, it will be added to your inbox.

1. Find the workflow in one of the workflow grids.
2. On the Action menu, choose **Acquire** to acquire the workflow without opening it or choose **Acquire and Open** to perform both actions.

11.7 Unacquiring a Workflow

This will unassign the workflow, remove it from your inbox, and return it to the workgroup's queue. This option is not available for roles that do not have a group inbox (queue).

1. Find the workflow in one of the workflow grids.
2. On the Action menu, choose **Unacquire**.

11.8 Assigning or Reassigning a Workflow

1. Find the workflow in one of the workflow grids.
2. On the Action menu, choose **Assign**.
3. Select the user to whom you want to assign the workflow.

   Note: Assign permissions are granted based on role and current workflow step.

11.9 Sorting the Inbox by Date

1. Select the desired inbox tab.
2. Click the **Received Date** column header to sort the inbox by the received date.
3. You can click the **Received Date** column header again to sort the inbox in reverse order.

11.10 Viewing the Workflow Process Diagram

The workflow process diagram shows all of the steps in the workflow and outlines the current step with a red border.

1. Open the workflow.
2. Click **View Process Diagram**.
3. To close the process diagram, click the Close button (x) in the upper right corner of the window.

11.11 Adding Time to a Workflow Step

Before you can complete a step in a workflow, you must enter the time spent on the step. The time spent and comment entered are used to generate workflow reports.

1. In the **Workflow Activities** section, click **Track Time** in the **Time Tracking** column.
2. In the dialog box, enter the hours that you have spent working on the step.
    - You can enter time in whole hours (e.g., 2) or with a single decimal (e.g., 2.6).
    - The time entered will override any previous time entered. You can view the previous time entered above the grid in the dialog box.
3. You can enter an optional comment. A comment is usually used to explain non-normal situations, such as why a step took longer than expected.
4. Click **Save**.

11.12 Adding Comments

1. Type comment text in the **Comments** box in the **Workflow Activities** section.
2. Click anywhere outside the box to save the changes.

11.13 Updating Comments

Only comments created in the current step can be edited.

1. Click in the **Comments** box.
2. Change the text as desired.
3. Click anywhere outside the box to save the changes.

11.14 Reviewing the Assembled File

For every review step, review the assembly:

1. Acquire the workflow from the workgroup queue (if necessary).
2. Open the workflow.

   The workflow defaults to the **Assembled File** Tab.
3. On the Action menu, choose **Open or Download** to review the file.

11.15 Processing Rejections

Rejection is the normal process to use when errors are discovered during review. Rejection sends the file back to the Editor for correction, and the review process restarts with the TD Lead. The best way to pass on corrections:

1. Download the review file.
2. Enter your markups and comments directly into the file.
3. Attach the file to the workflow.
4. Click **Reject**.

   A red "**R**" displays in the **Alert** column of the workflow screen.

The Quick Fix button is used for small fixes such as typos or minor grammatical corrections that do not affect the content. The file is sent back to the Editor for correction, but the workflow is returned to you so that it can continue relatively uninterrupted.

12 Introduction to CACTUS DITA Authoring

Darwin Information Typing Architecture (DITA) is an OASIS Standard that defines an XML architecture for designing, authoring, publishing, and managing content. In combination with the DITA Open Toolkit (an implementation of the information types defined in the DITA standard), DITA provides a powerful Open Source approach to structured authoring in XML. DITA and the Open Toolkit support content reuse, conditional publishing, and repurposing to different output formats.

The primary components of DITA are topics and maps. Topics are the primary building blocks of content. In addition to the core DITA topic types—concept, task, and reference—specialized topics and maps exist for learning and training applications. TD editors use this DITA learning and training framework with SyncRO Soft oXygen XML Author as the authoring tool, integrated with the CACTUS repository and publishing system. The TD manual types created are the Flight Training Manual (FTM) and Interactive Training Manual (ITM).

The overall process of authoring and publishing training content includes the following main activities:

- Developing a training strategy and design.
- Using the CACTUS workflow to initiate content development.
- Using oXygen XML Author to create training content in DITA format.
- Using the CACTUS publishing engine to assemble output.

13 Training Strategy and Design

This phase of training development includes planning for reuse, understanding the automated assignment of metadata in topics and maps, and being familiar with publishing options.

13.1 Reuse

One obvious benefit of reuse is avoiding the time and cost of creating duplicate, or near-duplicate, content. Another is that content can be changed in one place and the change will be reflected wherever the content is used. This avoids the costly and error-prone process of replacing all instances of the information. The result is cost avoidance and higher quality documentation. The advantages multiply the more times the information is revised.

What types of information can be reused? Consider product names, model numbers, procedures, and systems information. Also consider approved content that must remain the same in any publication, such as a copyright statement or branding elements. Identify repetitive content; it is a prime candidate for reuse. Larger blocks of content can also be reused, such as general descriptions, high-level processes, and subprocedures.

13.2 Metadata

Metadata (e.g., title, airplane model, reason for change) describes content. It is used for searching and sorting information.

When you begin a new file in CACTUS, you enter required metadata. A portion of that metadata is used to create the file name automatically, and other metadata elements are written into the file.

In DITA topics, metadata is included in the <prolog> element at the beginning of a topic. In maps, the <topicmeta> element is used to assign metadata. In book maps, <bookmeta> includes metadata that applies to all of the publication. By default, metadata assigned in the map overrides the equivalent metadata in the individual topics in a publication. Assigning this kind of metadata at a lower level within a topic can limit the ability to reuse the topic.

Note: See the

Special Characters appendix for a list of characters not allowed in property fields, such as title, keywords, and comments.

13.3 Publishing Requirements

The common training output formats are PDF (for the FTM) and EPUB 3 (for the ITM). The CACTUS system allows you to identify content that is specific to only one manual type.

14 Topics

The two primary components of DITA authoring are topics and maps. A well written topic covers a single subject and may be understood “stand-alone.” It does not require other topics to be conceptually complete.

TD editors use the DITA learning and training topic customizations (known in DITA as *specializations*). Authoring templates provide preconfigured starting points for content development. The templates are based on the learning and training topic specializations shown in Table 14-1.

Note: The general DITA glossary topic type is also used.

14.1 Learning and Training Topic Types

Table 14-1 DITA Learning and Training Topic Types

Topic Type	Used For
learningOverview	Learning objectives, prerequisites, duration, and so on.
learningContent	Core learning content such as Lesson Data, Lesson Plan, Modules, and Systems Knowledge.
learningAssessment	Questions or interactions that measure progress, encourage recollection, and reinforce the learning content.
learningSummary	Debrief section.

14.2 Topic-Based Authoring

Authoring information as stand-alone topics is central to the philosophy of DITA and critical to realizing its benefits. Topics support reuse because small information units can be recombined easily by using references in maps. To create chapters of a book, multiple topics should be combined by using the map, not by combining sections into a single file.

14.3 Elements and Attributes

Like all XML documents, DITA topics are composed of elements and attributes.

14.3.1 Elements

Elements consist of markers that define the start and end of content, as in HTML. In fact, several DITA elements are the same as their HTML counterparts. All element names begin with “<” and end with “>”. Each element consists of a start tag, such as <step>, and an end tag, such as </step>. Text entered between the start and end tags forms the element’s content.

14.3.2 Attributes

Attributes provide information about an element and its content. Attributes are defined in an element's start tag and consist of a name and a value (bounded by quotation marks). For example, *href* is an attribute of the *image* element: <image href="images/737_tr_unit.png"/>. The value of this attribute is *images/737_tr_unit.png*.

Attributes are commonly identified in documentation preceded by an "@" sign. For example, the href attribute is commonly represented as @**href**.

The most important attribute is @**id**. It is required on all topics and available on most elements. It is used to identify an element as unique and to reference the element from any other location. The @**id** attribute and value are automatically written into the file when you start a topic, map, or book map in CACTUS.

14.4 Authoring Templates

Although authors and editors of CACTUS training content sometimes use the DITA learning and training topic types directly, most authoring is performed by using preconfigured templates based on the DITA topic types.

Ten authoring templates are available, as shown in Table 14-2.

Table 14-2 DITA Authoring Templates

Training Manual Section	Authoring Template	DITA Topic Type Base
Syllabus	Syllabus.dita	learningContent
Overview	Lesson Overview.dita	learningOverview
Data	Lesson Data.dita	learningContent
Lesson Plan	Lesson Plan.dita	learningContent
Modules	Modules.dita	learningContent
Debrief	Lesson Debrief.dita	learningSummary
Procedures	Procedures.dita	learningContent
Systems Knowledge	Systems Knowledge.dita	learningContent
Self-Assessment	Assessment.dita	learningAssessment
Glossary	Glossary.dita	glossary

15 Maps

DITA maps organize and combine topics and other content into chapters, which in turn may be combined into other groupings such as volumes. They specify the order and hierarchy of topics and may be used to reference other maps and non-DITA resources, such as media files. Maps define the online navigation or table of contents for a deliverable. Map files have a *.ditamap* file name extension.

15.1 Basic Map Structure

Each topic has its own map, referred to as a "chapter" map. These chapter maps are then collected in a book map, as described in the next segment. The following chapter flowmap structure is recommended.

<map> (required)

<title> (required)

<topicmeta> (for metadata, required)

one or more <topicref> or <mapref> elements (required)

15.2 Book Maps

CACTUS training publications consist of multiple chapters. A map specialization known as a book map is used to combine chapters into training manuals.

A book map supports traditional book components such as front matter, chapters, and glossaries. Book maps also include metadata that defines overall information about the book, which may be used to generate content on the cover and title page. Maps are used to represent chapters in the manual. The book map contains references to the individual chapter maps.

The following is a recommended structure of a book map. Elements that are included should be used in the order shown.

<bookmap> (required)

<booktitle> (required)

<bookmeta> (for metadata, required)

<frontmatter> (required)

<booklists> (required)

<toc> (required)

one or more <part> elements (for volumes, optional)

one or more <chapter> elements (required)

any number of <appendix> elements (optional)

<backmatter> (optional)

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16 oXygen XML Author

oXygen XML Author is the DITA authoring tool. Before starting to create learning content, it is important to be familiar with these oXygen fundamentals:

- Configuring settings.
- Understanding the CACTUS and oXygen workflow.

16.1 Configuring oXygen

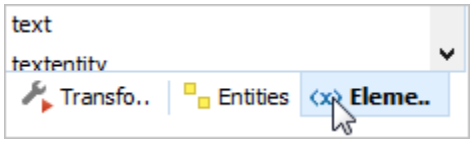
oXygen offers a rich set of user preferences and option settings. The settings in this segment are recommendations; authors may change them to suit their preferences.

16.1.1 Selecting Views

The panes displayed in the left and right columns of the oXygen window are known as *views*. To display a view, choose **Window** > **Show View**.

When you start oXygen, several views are displayed by default.

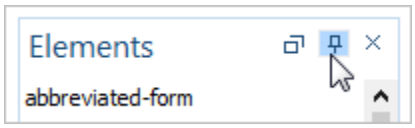
- You switch views by clicking their tabs at the bottoms of the columns:



- You close views by clicking the Close box in the upper right corner of the view pane:



- Pinning the view, by clicking the pushpin button until it points down, will keep the pane open and visible.

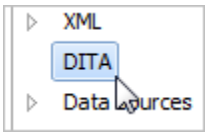


The recommended views: Attributes view above Elements view in the right column.

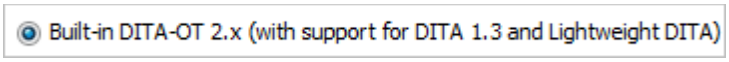
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16.1.2 Selecting the DITA Open Toolkit Version

The Preferences dialog box (**Options** > **Preferences**) contains dozens of configurable preferences. The most significant settings are on the DITA tab.

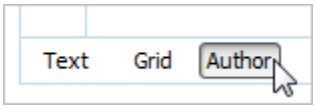


In the DITA Open Toolkit area, be sure that DITA OT 2.x is selected.



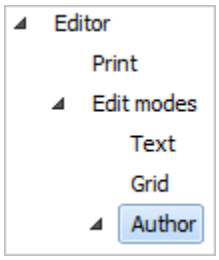
16.1.3 Selecting a Tags Display Mode

oXygen provides three main modes for working with DITA XML: Text, Grid, and Author. Most training authors and editors work in Author mode, which is similar to the WYSIWYG authoring view in Word. To display DITA content in Author mode, select **Author** at the bottom of the main oXygen window:

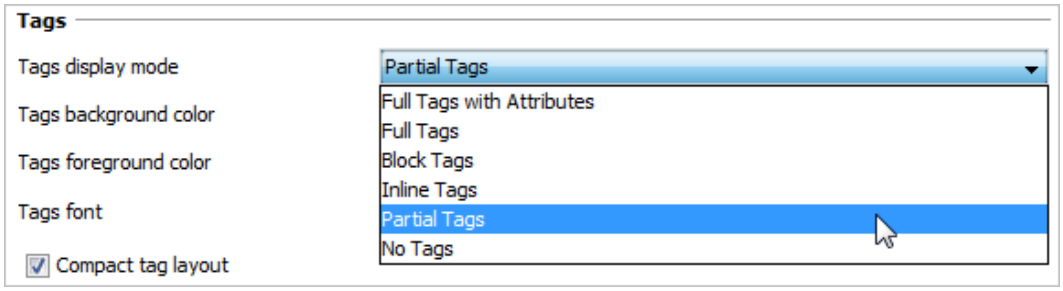


In Author mode, you can choose from a number of options for displaying the XML tags. To select a tags display mode in the Preferences dialog box:

1. Choose **Options** > **Preferences**.
2. In the **Preferences** dialog box, select **Editor** > **Edit modes** > **Author**.



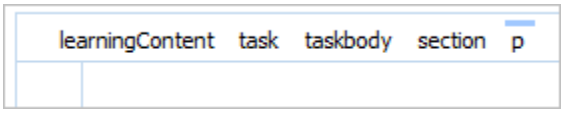
3. In the **Tags** area, you can select a tags display mode:



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Partial Tags mode is the default; the screen captures in this document show Partial Tags mode.

The Partial Tags mode includes a breadcrumb to identify the current position in the document. The breadcrumb is located at the top of the authoring window below the file name tab. Each node of the breadcrumb represents an element of DITA structure:



Note: Experience with using the breadcrumb is highly recommended. It is particularly useful for selecting only a specific element for copying or deletion.

16.1.4 Enabling Spell-Check

To enable automatic spell-check in oXygen:

1. On the oXygen menu bar, choose **Options** > **Preferences**.

   The **Preferences** dialog box is displayed.

2. Select **Editor** > **Spell Check**.

3. Select the **Automatic spell check** option.

4. Close the dialog box and restart oXygen.

16.2 CACTUS/oXygen Workflow

Initiating a file in CACTUS, editing in oXygen, and publishing through CACTUS follows this general workflow:

1. The TD lead creates a workflow from a book map and forwards the workflow to you through the Editor's queue.

2. If the workflow requires a new file, first create a "shell" for a new file in CACTUS, based on an authoring template or a DITA basic topic type.

   - Whenever possible, make a copy of a closely related existing file to start a new file, to reduce work steps.

   - Make note of the file name, you will need the name to identify the topic for referencing.

3. Locate and open the workflow.

4. Associate any required maps and topics with the workflow.

5. From the workflow window, check out any file that requires modification, including maps to which you may want to add new files.

6. Edit the files in oXygen.
7. Run on-demand validations.
8. From the workflow window, check in all files. (Check-in validations are triggered.)
9. Publish the files and review the result. (The file is automatically attached to the workflow on the **Assembled File** tab.)

When satisfied, forward the workflow for further reviews.

17 Creating and Saving Files in CACTUS

Topic, map, and book map files are initiated in CACTUS. For the procedures for initiating files and opening files to edit in oXygen, see Managing Files and Folders.

When you create a file, you are required to enter metadata. Required and optional metadata are shown in Table 17-1 through Table 17-3.

Table 17-1 Required and Optional CACTUS Topic Metadata

Required	Title Topic Type Model (prepopulated) Course Type (prepopulated) Localization Reason for Change
Optional	Regulatory Agency Customer Code Comments Keywords

Table 17-2 Required and Optional CACTUS Map Metadata

Required	Title Model (prepopulated) Course Type (prepopulated) Localization Reason for Change
Optional	Regulatory Agency Customer Code Comments Keywords

Table 17-3 Required and Optional CACTUS Book Map Metadata

Required	Document Number Title Model (prepopulated) Course Type (prepopulated) Course Number Regulatory Agency Customer Code Airport Info Localization Product Type Release Number Release Date Reason for Change
Optional	Minor Model Catalog Number Customer Catalog Number Comments Keywords

17.1 Saving Files

File names are generated in CACTUS as a concatenation of the CACTUS metadata. Example:

topic_lo_737_All_D_FAA__en-US_00002.dita

Note the filename so that you can refer to it later in maps, book maps, and workflows.

When you create a folder or file, it is saved in CACTUS in the folder that you had selected when you clicked the Create button on the menu bar.

17.2 Folder Structure

In the CACTUS repository, the following folder structure is used. It is displayed in the directory tree on the left side of the CACTUS window.

Training Development

Assembled Files/Bookmap/Map/Topic

Model

Differences

Other

Recurrent

Specialty

Type Rating, Transition

17.3 Check-In/Check-Out

To update files that you have saved and checked in, see the procedures for checking files out, and opening them to edit in oXygen, in Managing Files and Folders. Files that you have checked out are saved on your computer in this location:

C:\Users\<*userid*>\Documents\cactus

Note: You can change the metadata values of a file during check in. When the file is checked out in the future, it will include the new metadata values.

18 Common Procedures

Some CACTUS authoring procedures are common across training manuals:

- Inserting elements.
- Selecting elements by using the breadcrumb.
- Adding attributes.
- Inserting references (FCOM and CAMA).
- Inserting cross-references.
- Attaching and removing applicability.
- Validating links and content.
- Using revision marking.
- Inserting media.
- Creating image hotspots.
- Styling inline text.
- Specifying a document type.

It is strongly recommended that you became familiar with the procedures in this segment; several of these procedures are referenced repeatedly in the segments that follow.

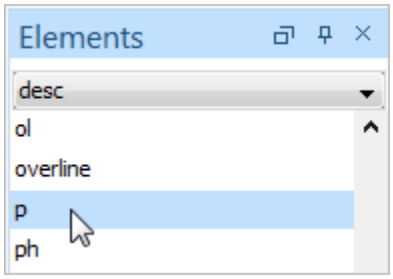
18.1 Inserting Elements

You can insert elements into a topic in two ways:

- The Elements view pane.
- The Content Completion Assistant.

18.1.1 Using the Elements View Pane

1. Click at the location in the topic where you want to add the element.
2. In the **Elements** view pane, double-click the element name:



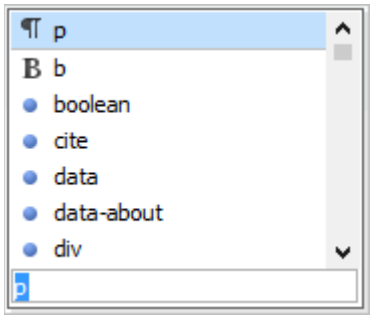
The element is inserted at the insertion point.

Note: If a view pane that you need is not displayed, choose **Window** > **Show View**. A list of all the oXygen panes will be displayed for selection in alphabetical order.

18.1.2 Using the Content Completion Assistant

1. Click at the location in the topic where you want to add the element. For example, click at the end of an existing element to add a new element.
2. Press Enter.

   The Content Completion Assistant is displayed showing the elements that are available in that context:

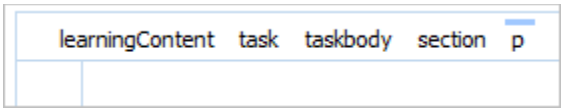


3. You can use a combination of text entry and the Up and Down Arrow keys to select an element, and then press Enter.

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18.2 Selecting an Element by Using the Breadcrumb

The Partial Tags mode in oXygen includes a breadcrumb to identify the current position in the document. The breadcrumb is located at the top of the authoring window below the file name tab. Each node of the breadcrumb represents an element of DITA structure:



You can select elements in the document by clicking the breadcrumb. This method of selecting is particularly useful for selecting only a specific element for copying or deletion.

18.3 Adding Attributes

You can add attributes to elements in two ways:

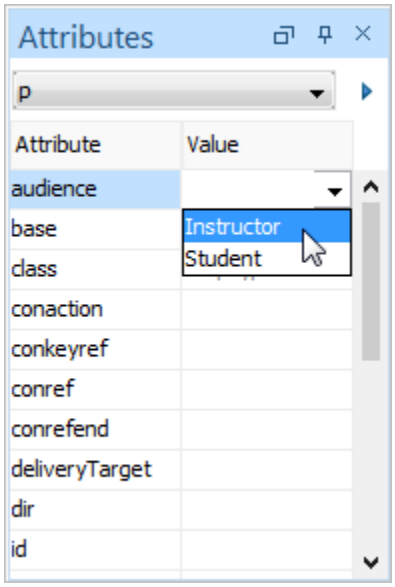
- The Attributes view pane.
- The in-place attributes editor.

One of the most common attributes to add is @**audience**, which allows you to create assembled output specifically for either instructors or students. The process of adding @**audience** and its value is the example shown in this segment.

18.3.1 Using the Attributes View Pane

1. By clicking or using the breadcrumb, select the element to which you want to add the attribute.

2. In the **Attributes** view pane in the upper right corner of the oXygen window, double-click in the Value cell to the right of the attribute name.



Note: If a view pane that you need is not displayed, choose **Window** > **Show View**. A list of all the oXygen panes will be displayed for selection in alphabetical order.

3. Type a value for the attribute, or select a value from the drop-down list.

   The attribute and its value are added to the element.

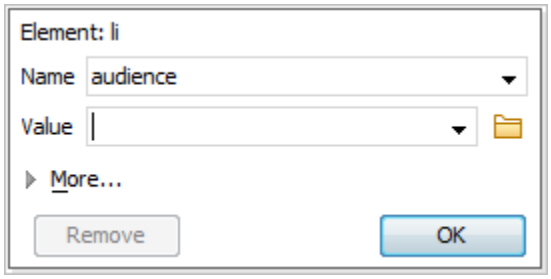
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18.3.2 Using the In-Place Attributes Editor

1. By clicking or using the breadcrumb, select the element to which you want to add the attribute.

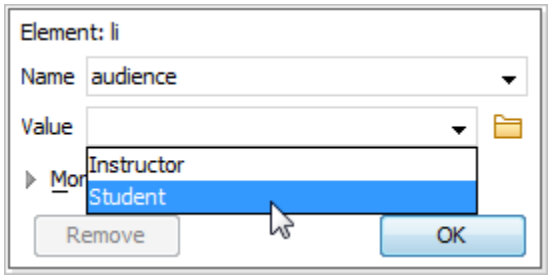
2. Press Alt+Enter, or right-click the element and choose **Edit Attributes** from the pop-up menu.

   An in-place attributes editor is displayed:



3. In the **Name** box, select an attribute from the drop-down list.

4. In the **Value** box, select or type the attribute value.



5. Click **OK** to close the attributes editor.

   The attribute and its value are added to the element.

18.4 Inserting References (CAMA and FCOM)

Citations to reference documents occur in all manuals. Use the **@camaId** or the **@fcomId** attributes to identify the type of reference.

1. Click anywhere in the element where you want to add a reference.

2. Depending on the document type to reference, add either the **@camaId** or the **@fcomId** attribute. See Adding Attributes.

3. In the attribute **Value** box, type the CAMA or FCOM ID value, and then press Enter.

   The ID is displayed above the element. CAMA IDs are indicated with a red square; FCOM IDs are indicated with a blue square.

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Note: If you need to insert more than one CAMA or FCOM value, separate the values with a comma.

18.5 Inserting Cross-Reference Links

To insert a cross-reference hyperlink to a location in the current topic or an external file:

1. Select the text that you want to link.
2. On the **CACTUS** menu, choose **Link** > **Cross-Reference**.
3. In the **Cross reference (xref)** dialog box, you can browse to an external target file in the repository or select a target element in the current document. If you select an external file in the repository, the list of target elements updates to show the elements in the external file.
4. Select a target element in the list, and then click **OK**.

   A cross-reference is inserted, and the text that you selected will be converted to a link when the file is transformed.

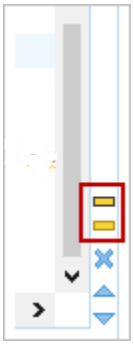
Note: If you want to remove the cross-reference, click at the beginning of the element where the cross-reference was inserted, and then press Backspace.



18.6 Validating Links and Content

If a hyperlink is invalid or broken, oXygen will automatically flag it within a few seconds of the link becoming broken, or within a few seconds of opening a file that contains invalid or broken links.

Yellow or red horizontal bars in the scroll bar indicate the location of broken links or other invalid content:

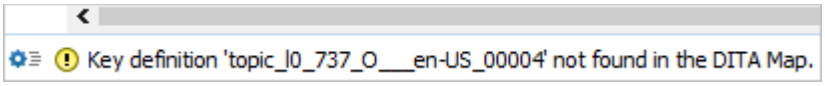


When you click on the bar, oXygen scrolls to the location and highlights it.

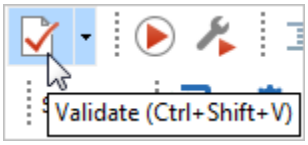
You can cycle through the bars by clicking the blue navigation arrows at the bottom of the scroll bar:



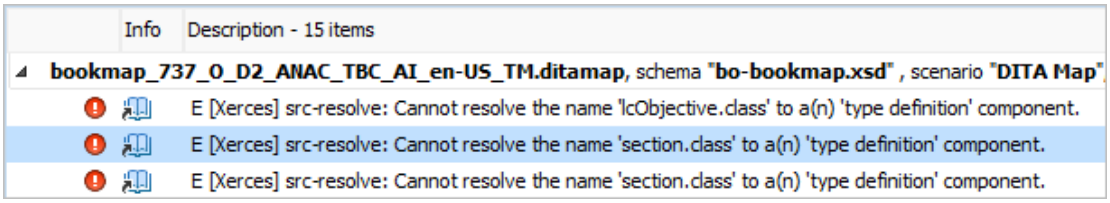
A message that describes the invalid content is displayed in the message area at the bottom of the oXygen editor panel:



To perform validation of all of the topic content on demand, click the **Validate** button on the toolbar or press Ctrl + Shift + V.



Validation messages are displayed in the Results view at the bottom of the oXygen editor panel:



18.7 Attaching and Removing Applicability

Applicability allows the author to mark content to create different versions of output, such as a customer-specific manual.

You add applicability to an element in the form of a Boolean expression. When the manual is assembled, only the elements with applicability that applies to the book map used in the assembly will be included. If an element does not have a Boolean, it will apply to all book maps.

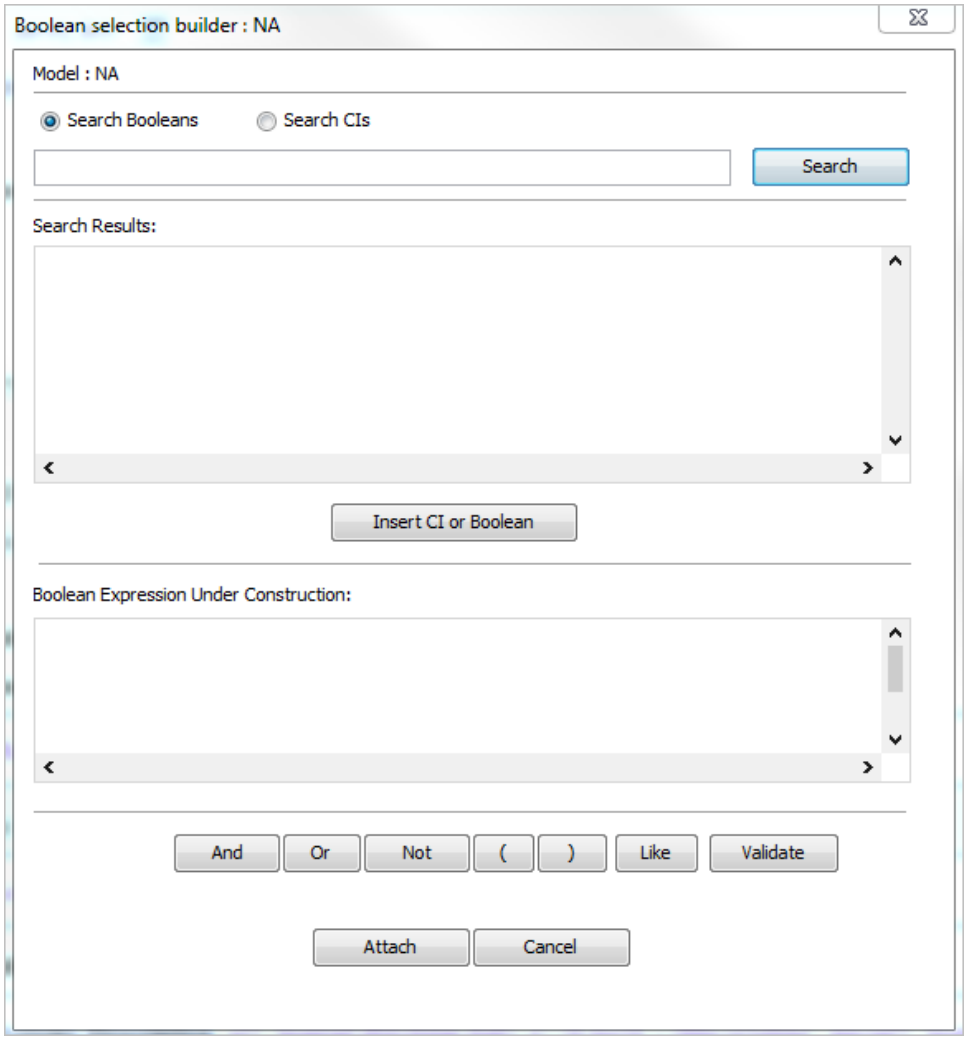
You can attach applicability to an element by selecting a preconfigured Boolean expression or building a custom expression from Configuration Items (CI).

Note: Terms in Boolean expressions are evaluated left-to-right, unless you enclose terms in parentheses. In that case, terms in parentheses are evaluated first.

To attach applicability to an element in a topic:

1. Click inside the element or just above it.
2. On the **CACTUS** menu, choose **Configure** > **Attach Applicability**, or press Ctrl+Shift+A.

   The **Boolean selection builder** dialog box is displayed.



   You can search for existing Boolean expressions or Configuration Items (CIs) or create a new Boolean expression.

- Search for existing Boolean expressions or CIs:

   a. Select **Search Booleans** or **Search CIs**.

   b. Enter a search term (e.g., **reg**) in the text box, and then click **Search**.

c. In the list of search results, select an item, and then click **Insert CI or Boolean**, or double-click the item.

   Your selection is displayed in the **Boolean Expression Under Construction** list.

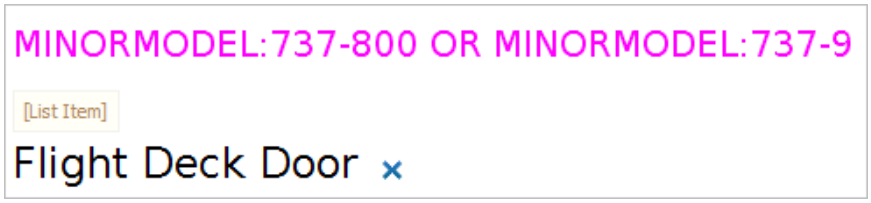
d. If the existing CI or Boolean expression needs to be edited, make the changes, and then click **Validate**.

   A message is displayed indicating whether or not the expression is valid; once validated, go to step e.

   Note: If the existing CI or Boolean expression does not need to be edited, you do not need to click **Validate**.

e. Click **Attach**.

   The expression is attached to the element and displayed above the element in magenta font color.



- Create a new Boolean expression:

  a. Build a Boolean expression using the operator buttons.



  b. When complete, click **Validate**.

  c. Once validated, go to step e.

To remove applicability that you have attached:

1. Click in the element to which you have assigned applicability.
2. On the **CACTUS** menu, choose **Configure** > **Detach Applicability**, or press Ctrl+Shift+D.
3. In the **Confirmation** dialog box, click **Yes**.

   The applicability is removed from the element.

18.8 Using Revision Marking

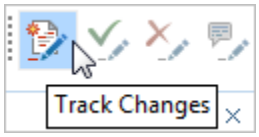
You can add revision marking when you add, modify, or delete content. The marking can include the following:

- Reason for the change.
- Highlighted background color.
- Vertical revision bar in the margin.

Three types of revision marking are supported: add, modify, and delete. Revision marking is added per change (i.e., you must configure and apply revision marking for each edit you want marked).

Note: Revision marking must be used and visible for all true revisions of training content. You may choose to hide marking for minor corrections, such as typographical edits. See Visibility.

Note: CACTUS revision marking is different from the oXygen Track Changes function:



Caution: Do *not* use oXygen's change tracking feature, as this will cause transformation errors. Use *only* CACTUS' change tracking feature.

Implementation

Revision marking for additions and modifications is implemented in the CACTUS function by adding attributes to the affected elements. Revision marking for deletions is implemented by replacing the content with an element, <deleted>, and an attribute, **@referencedElementID**, that references the content that was removed (which is retained in the CACTUS repository).

The revision applies on the element level. For example, if you make a partial change within a paragraph, revision marking is applied to the entire paragraph element.

Visibility

You can control whether the marking is visible or hidden. You may choose to hide marking for minor corrections of typographical errors or other changes that do not affect training content, to avoid interrupting the reading experience unnecessarily.

By default, revision marking is visible. This has the following effects in oXygen Author mode and on published output.

In Author mode in oXygen:

- The reason for the revision is displayed above the affected element.
- An element that is added or modified, and the reason for the revision, are highlighted with cyan background color.
- An element that is deleted, and the reasons for the deletion, are highlighted with red background color.

On output:

Note: An instructor guide may contain revision marking; a student guide should not.

- The reason for the revision is displayed in the summary of changes in the front of the manual, not at the location of the revision.
- Revised elements are marked with a vertical bar in the outside margin for FTM PDF outputs.
- An element that is added or modified, and the reason for the revision, are highlighted with cyan background color for FTM PDF outputs.
- An element that is deleted is identified by the word “Deleted” in red font color in the text (FTM) or with red background color (ITM).

Although revision marking is visible by default, you can change it to hidden for minor corrections. The effects of selecting visible or hidden revision marking are shown in Table 18-1.

Table 18-1 Effects of Visibility on Revision Marking in oXygen and on Output

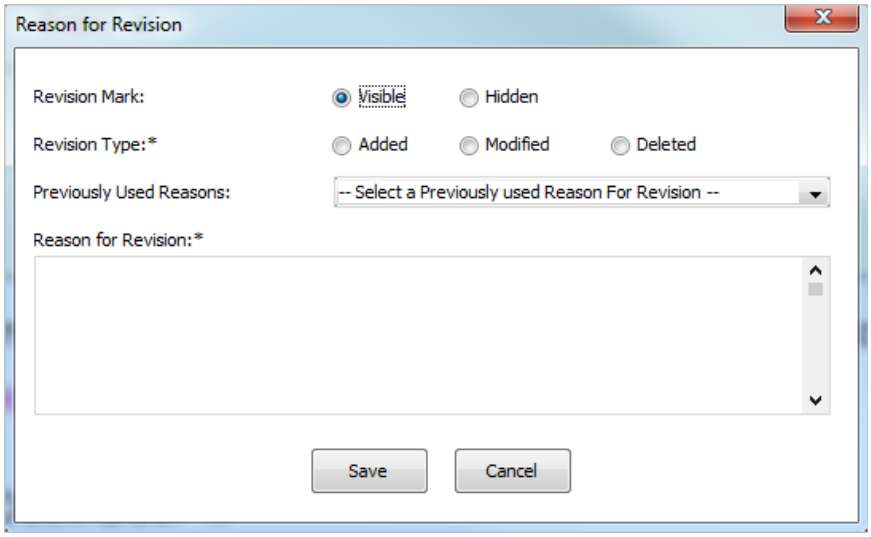
Visibility	Marking	oXygen Author Mode	Output
Visible	Reason for change	Displayed inline above element.	Displayed in summary of changes.
	Highlighting (ITM)	Displayed on reason for change and element.	For additions and modifications, displayed on element. For deletions, replaced with *Deleted* in red background highlighting in ITM.
	Revision bar (FTM)	Not displayed.	Displayed in outside margin in FTM.
Hidden	Reason for change	Displayed inline above element.	Displayed in summary of changes.
	Highlighting (ITM)	Not displayed.	Not displayed.
	Revision bar (FTM)	Not displayed.	Not displayed.

18.8.1 Adding Revision Marking

To add revision marking to an element in a topic:

1. Click anywhere on the line where you want to add revision marking.
2. On the **CACTUS** menu, choose **Configure** > **Insert Revision Marking**.

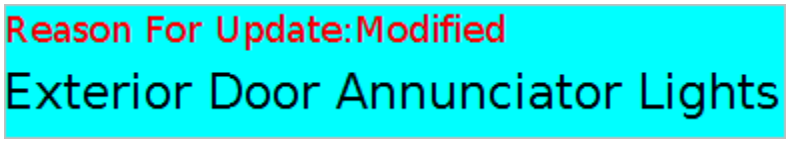
   The **Reason for Revision** dialog box is displayed.



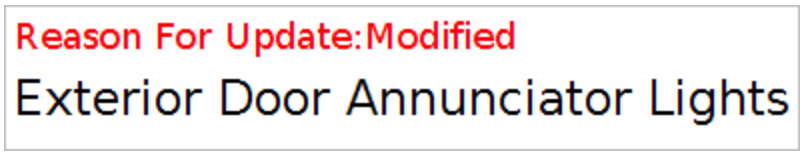
3. Select whether the revision marking is to be **Visible** or **Hidden**. For more information on the effects of these visibility options in oXygen and on output, see Table 18-1.
4. Select a revision type: **Added**, **Modified**, or **Deleted**. This selection is required.
5. In the **Reason for Revision** text box, enter the reason for the revision. If you have specified one or more reasons previously, you can select a reason from the **Previously Used Reasons** drop-down list; your selection will be entered into the text box.

   The reason for revision is required.
6. Click **Save**.

   The reason is inserted into your document above the affected element. If you have selected Visible revision marking, the content is highlighted.



   If you selected Hidden revision marking, only the reason is displayed.



18.8.2 Modifying Revision Marking

To modify revision marking that you have added:

1. Click anywhere on the line where you have added revision marking.
2. On the **CACTUS** menu, choose **Configure** > **Insert Revision Marking**.
3. Modify any of the selections you have specified, and then click **Save**.

   Your changes are applied to the revision marking.

18.8.3 Removing Revision Marking

To remove revision marking that you have added:

1. Click anywhere on the line where you have added revision marking.
2. On the **CACTUS** menu, choose **Configure** > **Detach Revision Marking**.
3. In the **Confirmation** dialog box, click **Yes**.

   The revision marking is removed from the element.

Note: You cannot restore deleted content by using Detach Revision Marking. You can either use Undo (Ctrl+Z) or copy and paste the content from an earlier version of the document.

18.9 Inserting Media

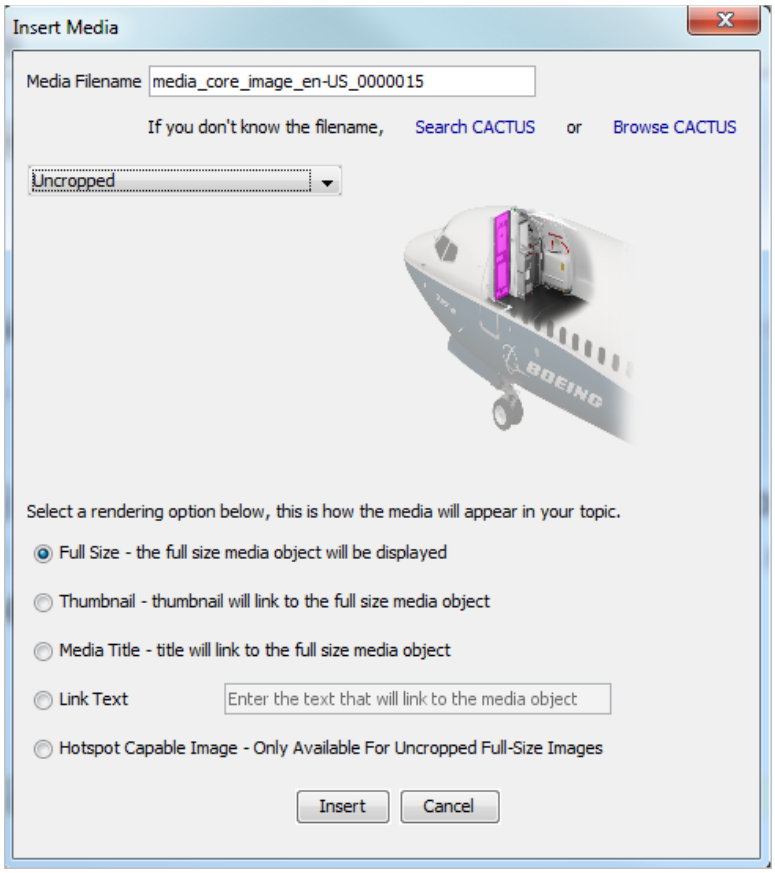
To insert media from the CACTUS repository:

1. Click where you want to insert the media object.
2. On the **CACTUS** menu, choose **Insert Object** > **Insert Media**.
3. In the **Insert Media** dialog box, type a file name and then press Enter, or click **Search CACTUS** or **Browse CACTUS**.
4. If you select **Browse CACTUS**, you can browse the directory tree. When you find the media file you want, click **Add**, then go to step 8.
5. If you select **Search CACTUS**, in the **Search Media** dialog box, from the **Media Type** drop-down list, select the type of file that you want to insert.

6. Enter any other search criteria, including **Model**, and then click **Search**.

7. In the list of files found, select a row, or press Ctrl+click to select multiple rows, and then click Insert.

   The **Insert Media** dialog box is displayed.



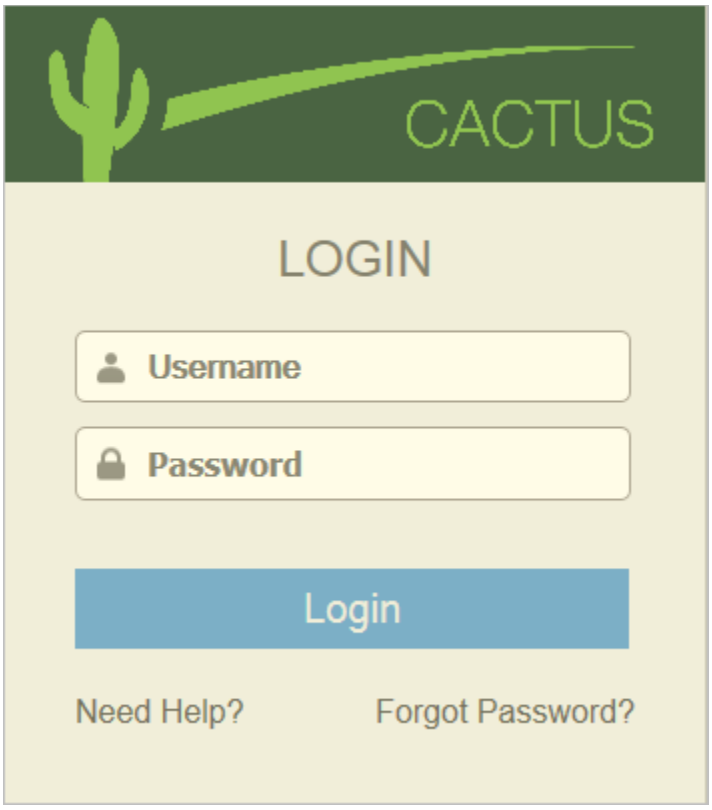
8. In the **Insert Media** dialog box, select whether you want to insert the image uncropped or cropped.

   Note: The default option is **Uncropped**. **Cropped** is only option if you have created a cropped version of the media object. See Cropping an Image for more information.

9. Select a rendering option.

   The screenshots below show examples of rendering options.

   - **Full Size** - the full size media object will be displayed.



- Thumbnail – thumbnail will link to the full size media object.



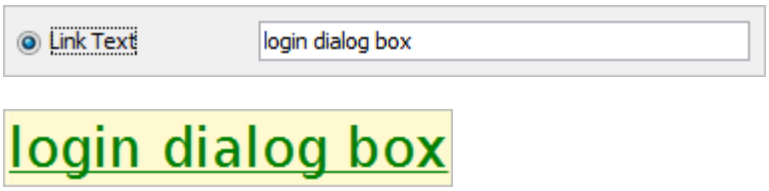
- Media Title – title will link to the full size media object.

  The title is used as the caption, which links to the media object. The caption is not editable in the document; the media object must checked out, and then the Title property modified.

  login

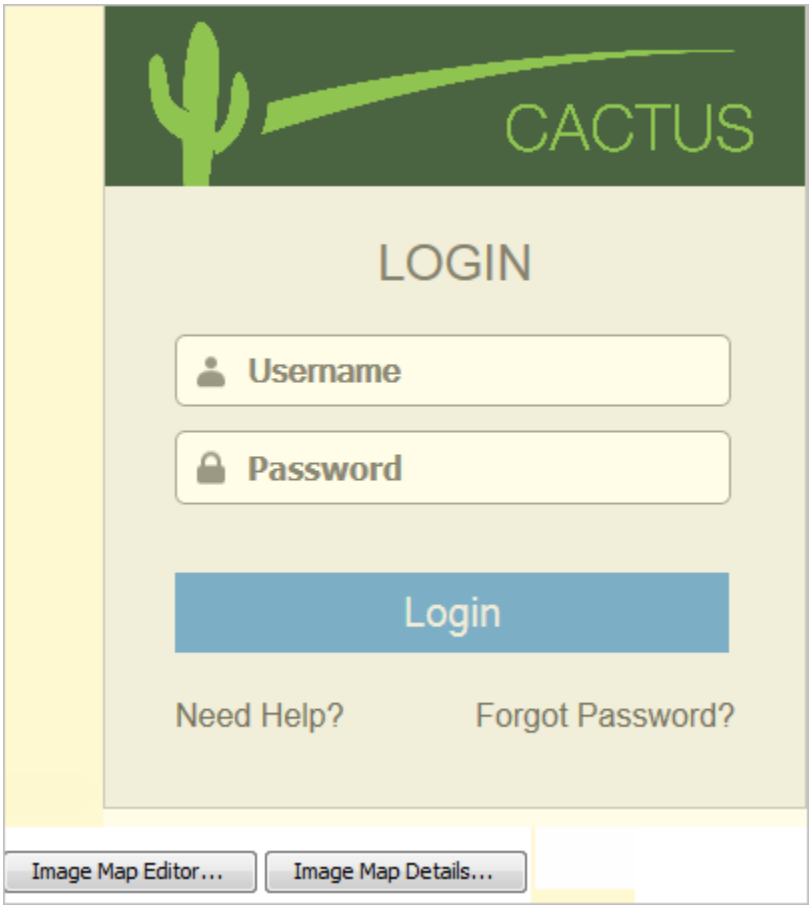
- Link Text.

  The text you entered in the **Link Text** field is inserted and linked to the media object.



- Hotspot Capable Image – Only Available For Uncropped Full-Size Images.

  The image is inserted with image map buttons that allow you to define a hotspot, and then view its details.



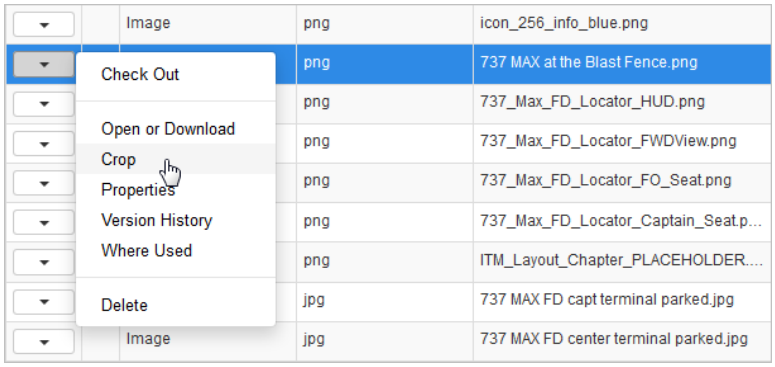
10. Click **Insert**.

    The media object is inserted into your document.

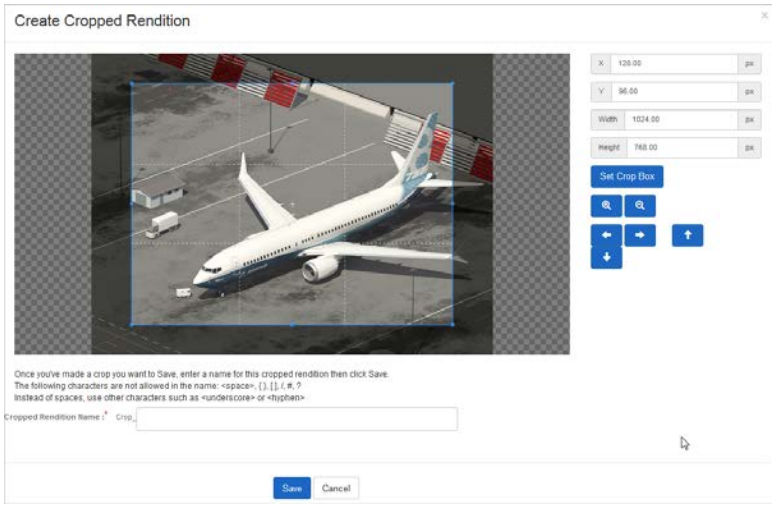
18.10 Cropping an Image

To crop an image, follow these steps:

1. Locate the image you want to crop.
2. On the Action menu for that image, choose **Crop**.



The **Create Cropped Rendition** dialog box is displayed.

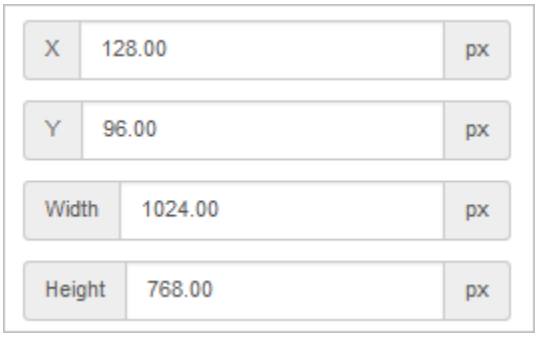


3. Crop the image using the following tools:

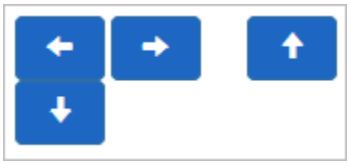
   - Resize the blue selection box around the image.



   - Manually enter values in the **X**, **Y**, **Width**, and **Height** fields.



- Click the arrows to move the image vertically and horizontally.



- Use zoom to adjust the magnification level.



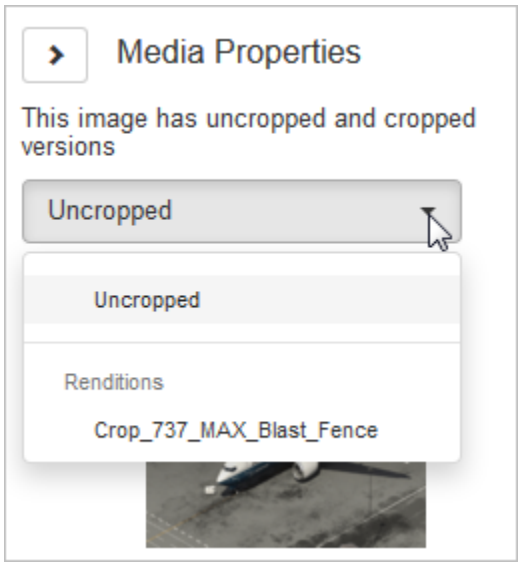
Note: As you crop the image using the blue selection box and arrows, the **X**, **Y**, **Width**, and **Height** fields automatically update to reflect the changing image size and location.

4. After you have completed cropping, click the **Set Crop Box** button above the zoom in/out buttons.

5. Enter a name for the cropped version of the image in the **Cropped Rendition Name** field, and then click **Save**.

   The cropped image is saved as an attachment to the original image.

   View the properties of the image you cropped, and you will see options to select the uncropped version or the rendition.



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18.11 Creating Image Hotspots (ITM Only)

Note: This function applies to the ITM only.

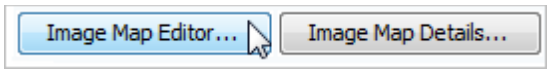
An image hotspot is an area in an image that contains a hyperlink.

Note: In CACTUS, hotspots can only link to Core Media images, video, and Captivate simulations (Sim).

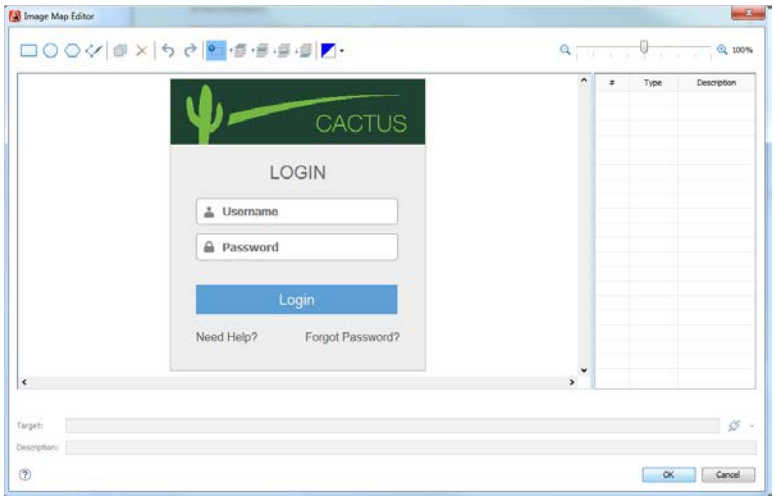
To create an image hotspot, follow these steps:

1. Insert the image, as described in Inserting Media, selecting the **Hotspot Capable Image – Only Available For Uncropped Full-Size Images** option.

2. Click the **Image Map Editor** button that is displayed below the inserted image.



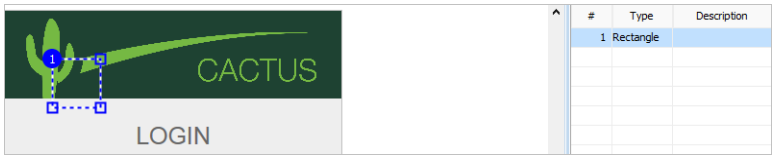
The **Image Map Editor** dialog box is displayed.



3. Select one of the shape options, left to right: **New rectangle**, **New circle**, **New polygon, Free drawing**.



The selected shape is dropped on the image and listed in the grid to the right.



Note: The shape is numbered and referenced in the grid.

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4. Drag and resize the shape to fit the area of the image where you want to create a hotspot.

   Note: You can select multiple hotspots.

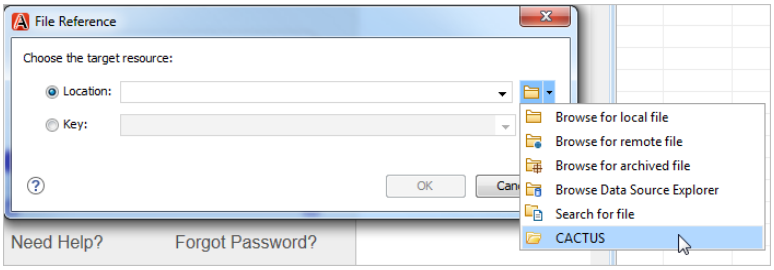
5. With the shape selected, as indicated by a blue dashed line, click the link icon, and then choose **File Reference**.



   Note: The link menu will not be enabled unless the shape is selected, as indicated by a blue dashed line.

   The **File Reference** dialog box is displayed.

6. In the **File Reference** dialog box, click the folder icon, and then choose **CACTUS**.



   Note: This feature works only with media stored in CACTUS.

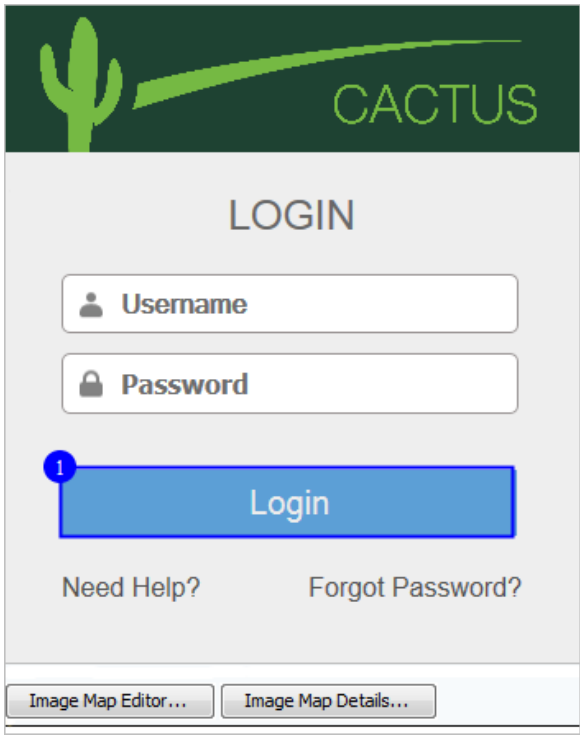
7. Browse the CACTUS directory to locate the file that you want the hotspot to link to, and then click **Add**.

   In the **File Reference** dialog box, the selected file is displayed in the **Target** field.

8. Click **OK**.

9. In the **Image Map Editor** dialog box, click **OK**.

A link is inserted to the target file that you selected, and the hotspot area is defined by a solid blue border.



Toggle **Image Map Details** to show the hotspot details or edit the shape/coordinates (i.e., location/size of hotspot). To edit the linked file, click **Image Map Editor**, select the hotspot (i.e., blue dashed line), and then edit the **Target** field.



10. If you added multiple hotspots, repeat the steps above, starting with step 5.
11. When you are finished creating hyperlinks, click **OK**.

18.12 Styling Inline Text

You may wish to apply specific styling to inline text to represent crew indications, or for other purposes such as emphasis or super- and subscripting.

18.12.1 Creating Indications

You may wish to apply specific styling to inline text to represent crew indications such as the following:

Audio Warning

Procedural Callout

Crew Information

Crew Attention

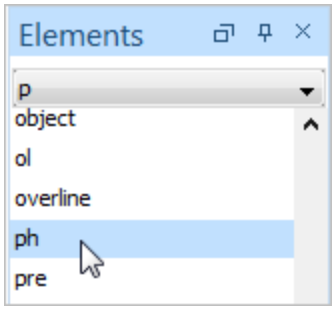
Crew Alert

FMC Active Waypoint

FMC Text

To apply indication styling to text:

1. Select the text to be modified.
2. In the **Elements** view pane, double-click **ph**.



   The selected text is wrapped with the <ph> element.

3. Click anywhere inside the **ph.** element.
4. Add the **@indications** attribute to the **ph** element. For more information on adding attributes, see Adding Attributes.

   For the value of the **@indications** attribute, in the **Value** box, select the indication type from the drop-down list.

5. Click **OK** to close the attributes editor.

   The indication formatting is applied to the selection.

18.12.2 Applying Subscript and Superscript

To apply Subscript or Superscript to inline text:

1. Select the text to be modified.
2. In the **Elements** view pane, double-click one of the following elements:

DITA element	To apply
sub	Subscript
sup	Superscript

18.13 Specifying a Document Type

You may wish to specify that some components should be included only in a single document type, either ITM or FTM. For example, you might want to restrict interactive media to ITMs only.

To specify that an element should be specific to a document type, add the **@deliveryTarget** attribute to the element with a value of either **ITM-ePub** or **FTM-PDF**. For more information on adding attributes to elements, see Adding Attributes.

19 Creating the Title Page

The training manual title page is automatically generated in CACTUS during assembly (publishing). The components such as the manual title, document number, revision number, and so on are assembled from the CACTUS metadata that you enter when you begin a book map.

The back of the FTM title page displays standard copyright information that is included unmodified in all manuals. (This information is located in the back of an ITM.) This information is contained in a single DITA topic file which resides in the CACTUS repository. The file is incorporated by a reference in the DITA book map.

20 Creating the Front Matter

The front matter of the training manuals includes the following parts:

- Revision History (in releases following the first).
- Table of contents.
- Syllabus.
- Crew Resource Management (CRM).

20.1 Revision History

The revision history occurs in both manual types, in all revisions following the original release. It includes changes since the last release.

The revision history is automatically generated by CACTUS from revision marking in the manual. It shows the type of revision (addition, modification, or deletion) and the reason for the change.

For more information on revision marking, see Using Revision Marking.

20.2 Table of Contents

The Table of Contents (TOC) is automatically generated by CACTUS during assembly.

20.3 Syllabus

The Syllabus is implemented by using the **Syllabus** template, which is based on the learningContent DITA topic type. The template is divided into separate sections for the FTM and the ITM because the Syllabus is authored differently in the two document types.

To create the syllabus (FTM or ITM):

1. From the **Choose Topic** drop-down list, choose the **Syllabus** template.
2. Once created, attach the Syllabus file to the appropriate workflow, and then check out the file.

20.3.1 Creating the FTM Syllabus

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Day Lists

The **DAY 1** heading has been provided in the template in <section><title>. The first subheading of DAY 1, **Classroom**, has been provided in the template in <section><p>.

1. Retain the subheading text or type over it as appropriate.
2. Type over the **Item** placeholders to create the Syllabus list.
3. To add new items, at the end of the entry, press Enter and then double-click **New sli** in the Content Completion Assistant.
4. Copy, paste, and revise these elements to build the Syllabus list for all days.
5. Delete the rest of the sections and placeholder text, which are specific to the ITM:

20.3.2 Creating the ITM Syllabus (ITM Only)

The top of the template is provided for use only in the Creating the FTM Syllabus section. Begin by deleting this section, and then follow the instructions below.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Day Lists

1. In each of the specialized sections (<section_introduction>, for example), type the heading in <title>.

   This heading will be formatted with the corresponding colored background on output.
2. Type over the Module and Profile placeholder **Text** to create Syllabus entries.
3. To add new entries, at the end of the entry, press Enter and then double-click **New p** in the Content Completion Assistant.
4. Delete unused sections and placeholder text.

20.4 Crew Resource Management (CRM)

The CRM section is standard information that is automatically included unmodified in all manuals. This information is contained in a single DITA topic file which resides in the CACTUS repository. The file is incorporated by a reference in the DITA book map.

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21 Creating the Introduction

The Introduction is implemented by using the learningContent DITA topic type.

> **Note:** The Introduction is similar across training manuals. It may be more efficient to copy an existing Introduction and edit it, rather than create a new one. See Copying a File if you want to use an existing file; otherwise, follow the steps below.

To create the introduction:

1. From the **Choose Topic** drop-down list, choose the **Learning Content** basic topic type.

2. Once created, attach the Introduction file to the appropriate workflow, and then check out the file.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

> **Note:** The DITA Short Description element is not used; it will not display on output. If you prefer, you may click inside the **Short Description** label and then press Delete to remove it.

Section

To insert a section and sectional heading:

1. Click at **learningContentbody**.

2. In the **Elements** view pane in the lower right corner of the oXygen window, double-click **section**.

3. Type over the **title** placeholder to create a sectional heading.

Paragraph

To create paragraphs:

1. Type over the **p** placeholder.

2. To create another paragraph, press Enter and then double-click **New p** in the Content Completion Assistant.

Bulleted List

To insert a bulleted list:

1. Press Enter at the end of a paragraph and then double-click **ul** in the Content Completion Assistant.

2. At the **li** placeholder, type the list item content.

3. To insert another list item, press Enter and then double-click **New li** in the Content Completion Assistant.

Student Note

To insert a student note:

1. Press Enter at the end of a paragraph and then double-click **note** in the Content Completion Assistant.

2. Following the **Note** label, type the content of the note.

   Use the **@type** attribute to change a standard NOTE to a CAUTION or WARNING. This feature is available for both the ITM ePub and FTM PDF outputs.

   Note: Only the “blank” (for simple NOTE), CAUTION, and WARNING @type selections are implemented in CACTUS.

22 Creating Lessons

The core of a training manual consists of three main parts:

- Lessons (including Lesson Overview, Lesson Data, Lesson Plan, and Debrief).
- Procedures (ITM Only).
- Systems Knowledge (ITM Only).

A self-assessment occurs at the end of each training module. An overall evaluation is provided at the end of each course.

22.1 Creating a Lesson Overview

The overview section is relatively brief and may contain some or all of the following:

- Lesson Summary (or Overview)
- Duration
- Prerequisites
- Lesson Objectives (or Objectives)
- Assessment and Minimum Proficiency

To create the lesson overview:

1. From the **Choose Topic** drop-down list, choose the **Lesson Overview** template.
2. Once created, attach the Lesson Overview file to the appropriate workflow, and then check out the file.

22.1.1 Overview Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

22.1.2 Lesson Summary

The Lesson Summary is implemented with the <lcIntro> element.

Heading

The standard Lesson Summary heading has been entered into the template.

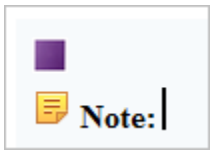
Paragraph

In <lcIntro><p>, type over the **Text** placeholder to create the summary paragraph.



Instructor Note

In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.



Bulleted List

Type over the **Bullet item** placeholders in the topic template.

Student Note

To insert a student note:

1. Press Enter at the end of a paragraph and then double-click **note** in the Content Completion Assistant.

2. Following the **Note** label, type the content of the note.

   Use the **@type** attribute to change a standard NOTE to a CAUTION or WARNING. This feature is available for both the ITM ePub and FTM PDF outputs.

   Note: Only the "blank" (for simple NOTE), CAUTION, and WARNING @type selections are implemented in CACTUS.

22.1.3 Duration

Duration is implemented with the <lcDuration> element.

Heading

The standard Duration heading has been entered into the template.

lcTime

In the <lcTime> element, following the **Time** attribute box, edit the text to specify the durations in hours and minutes. End each duration with a period.

Note: Any text in the **Time** box will be ignored in the output.

22.1.4 Prerequisites

The Prerequisites section is implemented with the <lcPrereqs> element. If your manual contains prerequisites, follow the steps below. If it does not, use the breadcrumbs to select **lcPrereqs** and then delete the section.

Heading

The standard Prerequisites heading has been entered into the template.

Prerequisites List

Type over the **Prerequisite** placeholders in the template.

22.1.5 Lesson Objectives

Lesson Objectives is implemented with the <lcObjectivesGroup><lcObjective> element.

Heading

The standard Lesson Objectives heading has been entered into the template.

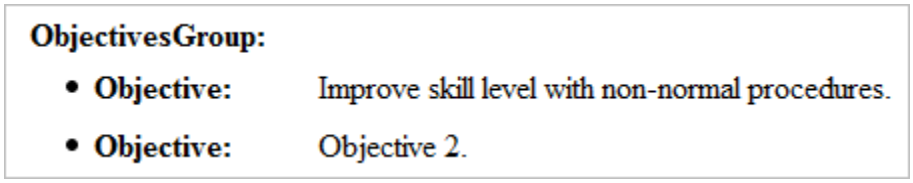
Objectives Stem

If desired, in <lcObjectivesStem>, following the **ObjectivesStem** label, type a clause or sentence introducing the objectives. If an objectives stem is not desired, click the **ObjectivesStem** label, and then press Delete to delete it.

lcObjectivesGroup

The <lcObjectivesGroup> is a container for 1 to *n* <lcObjective> elements, which render as a bulleted list.

1. Following each **Objective** label, type a learning objective over the placeholder values in the list.



2. If you need to create sub-objectives, type them over the **Sub-objectives** placeholder text.

   Note: At this time, only first and second level objectives are available.

3. To create more objectives list items, on the last item, press Enter, and then double-click **New lcObjective** in the Content Completion Assistant.

   To create more sub-objectives, on the last item, press Enter, and then double-click **New lcObjective_Sub** in the Content Completion Assistant.

4. Delete any unused <lcObjective> elements.

22.1.6 Assessment and Minimum Proficiency

This content unit is implemented inside a <section> element.

Title

The standard Assessment and Minimum Proficiency title has been entered into the template.

Paragraph

In <section><p>, type over the **Text** placeholder to create a one-paragraph description of proficiency after completing the lesson.

22.2 Creating the Lesson Data

The Lesson Data section provides the assumptions for the lesson. It contains the following four subsections:

- Route of Flight.
- Performance.
- Weather.
- Clearance.

A Flight Plan subsection is also included for 717 training.

To create the lesson data:

1. From the **Choose Topic** drop-down list, choose the **Lesson Data** template.
2. Once created, attach the Lesson Data file to the appropriate workflow, and then check out the file.

22.2.1 Lesson Data Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

22.2.2 Adding Rows to Lesson Data Tables

The tables in the Lesson Data template are preconfigured for your use. It is unlikely that you will need to add rows to the tables. If you do have a requirement to add rows, follow these steps:

1. Click in the last row of the table.
2. On the **CACTUS** menu, choose **Table** > **Insert Row Below**.

   A new row is inserted.
3. Repeat steps above as required.

To add boldface styling to row labels (e.g., ORIGIN, RUNWAY) in the tables:

1. Select the label in the left column.
2. In the **Elements** view pane, double-click **b**.

   Boldface styling is applied to the label.

22.2.3 Route of Flight

In CACTUS, from the **Choose Topic** drop-down list, choose the **Lesson Data** template.

Use the preconfigured Route of Flight table for all models except 717. A Flight Plan table is provided for use in 717 training only.

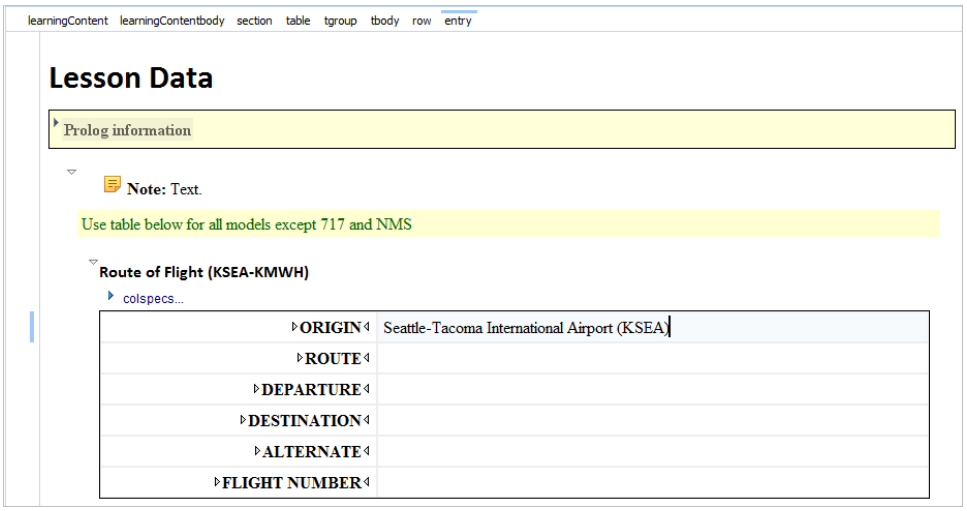
Note: Delete the 717 Flight Plan table if it is not used.

Heading

The standard Route of Flight heading has been entered into the template. Following the title, type the route in parentheses; for example, **(KSEA-KMWH)**.

Route Table

The standard labels have been entered into the table in the left column. Enter the corresponding values in the right column In the <entry> elements.



22.2.4 Performance

Use the **Lesson Data** template.

Use the preconfigured Performance table specific to the airplane model.

Note: Delete the unused Performance tables.

Heading

The standard Performance heading has been entered into the template.

Performance Table

The standard labels have been entered into the table in the left column. Enter the corresponding values in the right column In the <entry> elements.

22.2.5 Weather

Use the preconfigured Weather table specific to the airplane model.

Note: Delete the unused Weather tables.

Heading

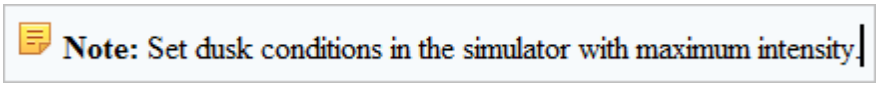
The standard Weather heading has been entered into the template.

Weather Table

For the 737CL, 747, 757, 767, 777, and 787 models, enter the labels into the table in the left column. The text that you enter will be bounded by . Tables with standard labels entered in the left column have been provided for the 737NG, 737 MAX, and 717. Enter the corresponding values in the right column in the <entry> elements.

Instructor Note

If an instructor note is required, in addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, @**audience="Instructor"**. Once tagged with @**audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.



If no note is required, use the breadcrumbs to select **lcInstructornote2** and then delete it.

Paragraph

If one or more paragraphs are required, type over the **Text** placeholder. If no paragraph is required, use the breadcrumbs to select **p** and then delete it.

22.2.6 Clearance

Use the **Lesson Data** template.

Heading

The standard Clearance heading has been entered into the template.

Paragraph

In <section><p>, type a single paragraph of clearance communication over the **Text** placeholder.

22.3 Creating the Modules Section (ITM Only)

Note: This section occurs in the ITM only.

The individual numbered Modules described below are implemented as separate topics. Specialized heading elements have been preconfigured in the template.

To create the modules:

1. From the **Choose Topic** drop-down list, choose the **Modules** template.
2. Once created, attach the Modules file to the appropriate workflow, and then check out the file.

22.3.1 Modules Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. Example:

Module 1 – Captains Preflight Procedure / 1:30 (1:45)

Headings

The standard section headings have been entered into the template. Styling and color are applied on output based on the "module_*abc*" heading element used.

Bulleted List

Type over the **Bullet item** and **Sub-bullet item** placeholders in the topic template.

22.4 Creating the Lesson Plan Section

The Lesson Plan section provides the tasks that the learner will perform. It is structured by the flight profile segments illustrated in the profile images at beginning of the section. It contains steps, notes, and reference citations.

The Lesson Plan section is implemented as separate topics and output as one content unit by using nesting and **@chunk** in the map. For more information, see Creating Maps.

To create the lesson plan:

1. From the **Choose Topic** drop-down list, choose the **Lesson Plan** template.
2. Once created, attach the Lesson Plan file to the appropriate workflow, and then check out the file.

22.4.1 Lesson Plan Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

22.4.2 Flight Profile Images

In CACTUS, from the **Choose Topic** drop-down list, choose the **Lesson Plan** template.

Use only the following elements:

<title>

<learningContent body>

<section>

Note: Delete the rest of the template content inside the <section> element.

Create a separate topic for each flight profile image.

Title

The topic title is created automatically from the Title that you enter when you create a topic in CACTUS.

Figure

Add the profile image by using the **Insert Media** command on the **CACTUS** menu. See Inserting Media for information on inserting an image.

22.4.3 Track Image

If your manual includes a track image, in CACTUS, from the **Choose Topic** drop-down list, choose the **Lesson Plan** template.

Use only the following elements:

<title>

<learningContent body>

<section>

Note: Delete the rest of the template content inside the <section> element.

Create a separate topic for each track image.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Figure

Add the track image by using the **Insert Media** command on the **CACTUS** menu. See Inserting Media for information on inserting an image.

22.4.4 Track Data

If your manual includes track data, in CACTUS, from the **Choose Topic** drop-down list, choose the **Lesson Plan** template.

Use only the following elements:

<title>

<learningContent body>

<section><title>

<sl><sli>

<lcInstructornote2>

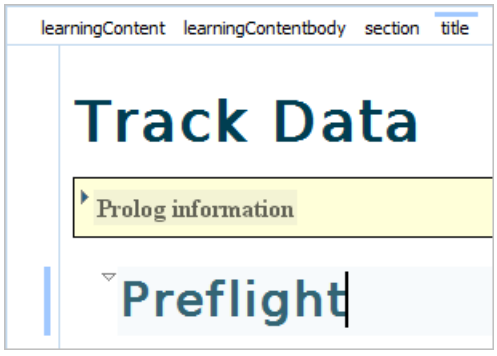
Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

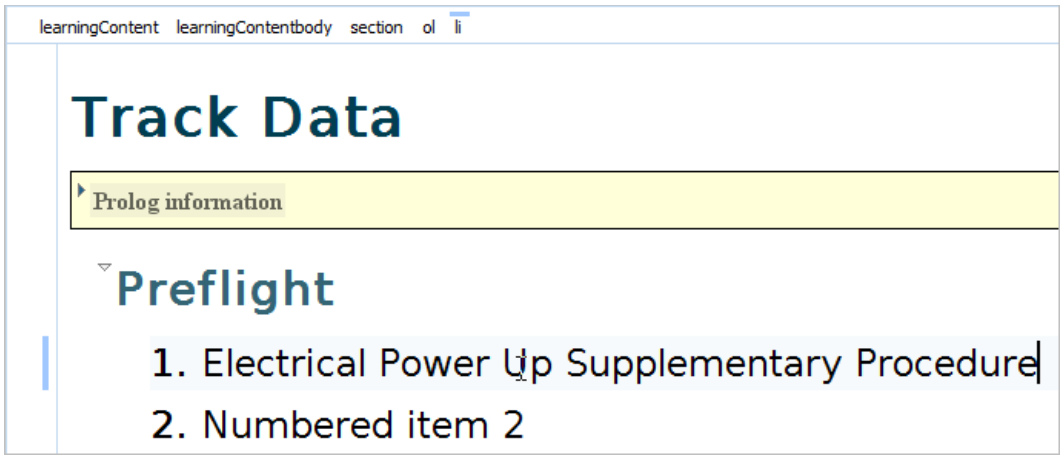
Phases of Flight

Create all phases of flight as separate <section> elements in a single topic.

1. In <section><title>, type the phase of flight heading (**Preflight**, for example) over the **title** placeholder.



2. Type the phase of flight entries over the **Numbered item** placeholder text. Number these entries manually.



3. If instructor notes are required, in addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.

   If notes are not required, use the breadcrumbs to select **lcInstructornote2**, and then delete it.

4. Repeat steps above as required to create all Track Data phase of flight blocks.

22.4.5 Device Familiarization

If your manual includes a device familiarization unit, in CACTUS, from the **Choose Topic** drop-down list, choose the **Lesson Plan** template.

Use only the following elements:

<title>

<learningContent body>

<section><title>

<lcInstructornote2>

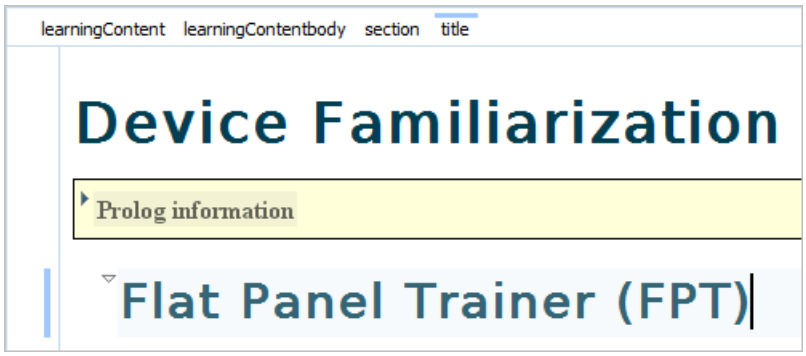
If your manual contains multiple device familiarization units, create one topic per unit.

Title

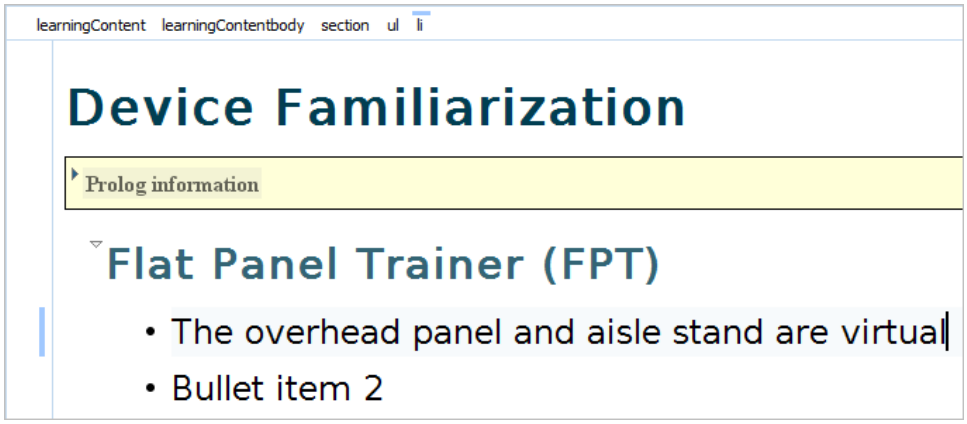
The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Device Topics

1. In <section><title>, type the device type heading at the **title** placeholder.



2. In , type the device topics over the **Bullet item** placeholder text.



3. If instructor notes are required, in addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.

   If notes are not required, use the breadcrumbs to select **lcInstructornote2**, and then delete it.

22.4.6 Daily Preparation Sections

In CACTUS, from the **Choose Topic** drop-down list, choose the **Lesson Plan** template.

Create each Daily Preparation segment (e.g., **Daily Preparation – Ground Ops**) as a separate topic. The sections are output as one content unit by using nesting and @chunk in the map under **Lesson Data**. For more information, see Creating Maps.

Create each Phase of Flight as a section (e.g., **TAXI**, **TAKEOFF**) inside the related **Daily Preparation** segment.

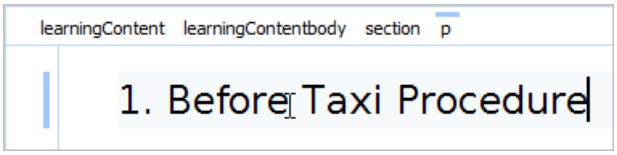
Use the elements of the template as required. Copy and paste elements to create additional sections as required. Delete unused elements.

Title

The title of each Daily Preparation segment (e.g., **Daily Preparation – Ground Ops**) is created automatically from the Title metadata that you enter when you begin a topic in CACTUS.

Step

Inside <section><p>, type the step content over the placeholder text. Three numbered steps with placeholder text are provided in the template.



Note: Number the steps sequentially across subsections. For example, if the steps in the TAXI subsection end with step 3, begin the next, TAKEOFF, section with step 4.

Bulleted List

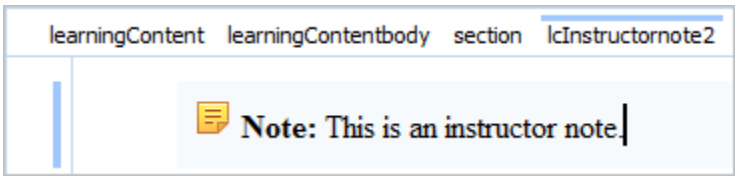
1. Type the list item content over the **Bullet item** placeholder text.

   Note: The bullet item <ul> is nested inside the <p> element. This causes the bulleted list to be indented as desired under the step text on output.

2. To create more list items, on the last item, press Enter, and then choose **New li** from the pop-up menu.

Instructor Note

In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.



22.5 Creating the Debrief

The Debrief section provides space for student and instructor evaluation of the lesson, a summary of the next lesson, and optional questions for students to consider. In addition to the section title, it contains headings, notes, and bulleted lists.

Note: Debriefs are standardized. Verify that the required version of Debrief is not already in CACTUS before creating a new one. If it already exists, link to the one you need instead; otherwise follow the steps below.

To create the debrief:

1. From the **Choose Topic** drop-down list, choose the **Lesson Debrief** template.

2. Once created, attach the Lesson Debrief file to the appropriate workflow, and then check out the file.

Debrief Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Summary Label

Ignore the **Summary:** label; it will not be displayed on output.

Headings

The standard headings (“Student Evaluation of Session,” and so on) have been entered into the template. They are implemented as <p> elements bounded by , which enables bold styling in oXygen. An **@outputclass** attribute has been added to enable bold styling on output.

First-Level Bulleted List

Inside , type the list item content over the **Bullet item** placeholder text.

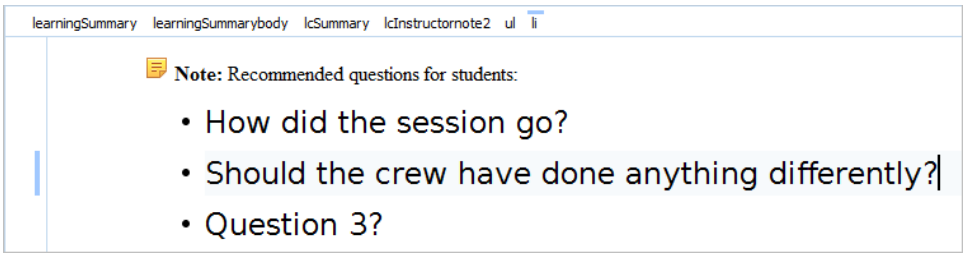
Instructor Note

In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience=“Instructor”**. Once tagged with **@audience=“Instructor”**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.

Second-Level Bulleted List

Second-level bulleted list items are questions that occur below notes. They are contained within the <lcInstructornote2> element.

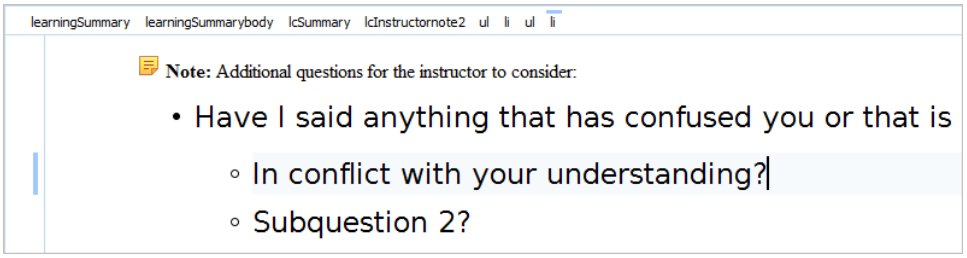
In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.



Third-Level Bulleted List

Third-level bulleted list items are questions that occur below bulleted questions. They are contained within the second-level bulleted lists below the <lcInstructornote2> element.

Inside <lcInstructornote2>
, type the list item content over the **Subquestion** placeholder text.



22.6 Creating the Evaluation

The overall training evaluation consists of four parts:

- Evaluation Overview.
- Evaluation Data.
- Evaluation Profile.
- Maneuvers List.

Four templates are required to complete the Evaluation unit:

- Lesson Overview template (for Evaluation Overview).
- Lesson Data template (for Evaluation Data).
- Lesson Plan template (for Evaluation Profile).
- Learning Content basic topic type (for Maneuvers List).

To create the evaluation:

1. From the **Choose Topic** drop-down list, choose the appropriate evaluation template listed above.
2. Once created, attach the Evaluation file to the appropriate workflow, and then check out the file.

22.6.1 Evaluation Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

The titles of unit topics (Evaluation Overview, Evaluation Data, etc.) are created automatically from the Title metadata that you enter when you begin a topic in CACTUS.

22.6.2 Evaluation Overview

Using the **Lesson Overview** template, follow the procedures in Creating a Lesson Overview.

22.6.3 Evaluation Data

Using the **Lesson Data** template, follow the procedures in Creating the Lesson Data .

22.6.4 Evaluation Profile

Using the **Lesson Plan** template, follow the procedures in

Flight Profile Images.

22.6.5 Maneuvers List

To create the maneuvers list:

1. From the **Choose Topic** drop-down list, choose the **Learning Content** basic topic type.

2. Once created, attach the Lesson Overview file to the appropriate workflow, and then check out the file. The Maneuvers List is implemented by using the Learning Content basic topic type.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Note: The DITA Short Description element is not used; it will not display on output. If you prefer, you may click inside the **Short Description** label and then press Delete to remove it.

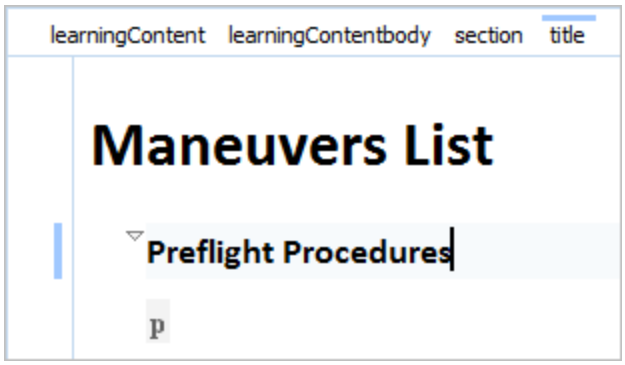
List of Maneuvers

1. Click at **learningContentbody**.

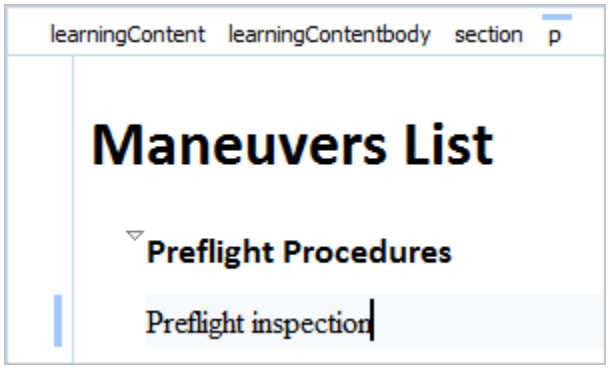


2. In the **Elements** view pane in the lower right corner of the oXygen window, double-click **section**.

3. Type over the **title** placeholder to create the maneuver section heading; for example, **Preflight Procedures**.

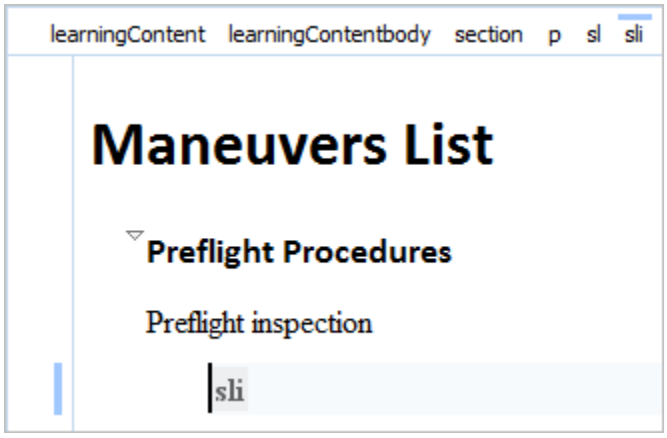


4. Type over the **p** placeholder to create a main entry in the maneuvers list.

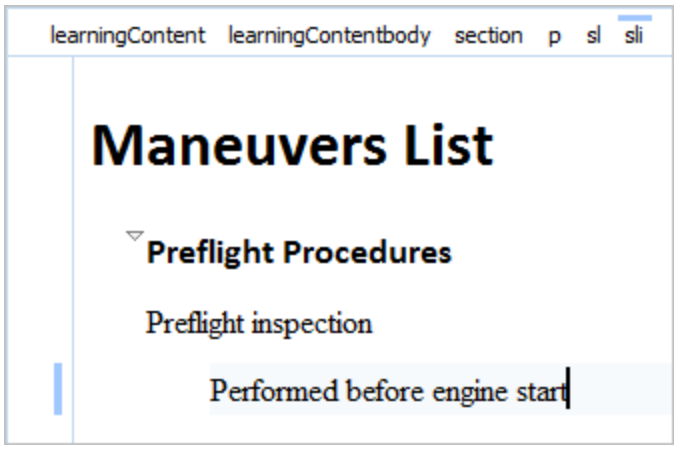


5. Main entries in the maneuvers list may have subentries. You enter subentries as DITA simple list items. Click at the end of the main entry that you typed, and then, in the **Elements** view pane, double-click **sl**.

   The **sli** (simple list item) placeholder is displayed. (The sl container is inserted in the background.)



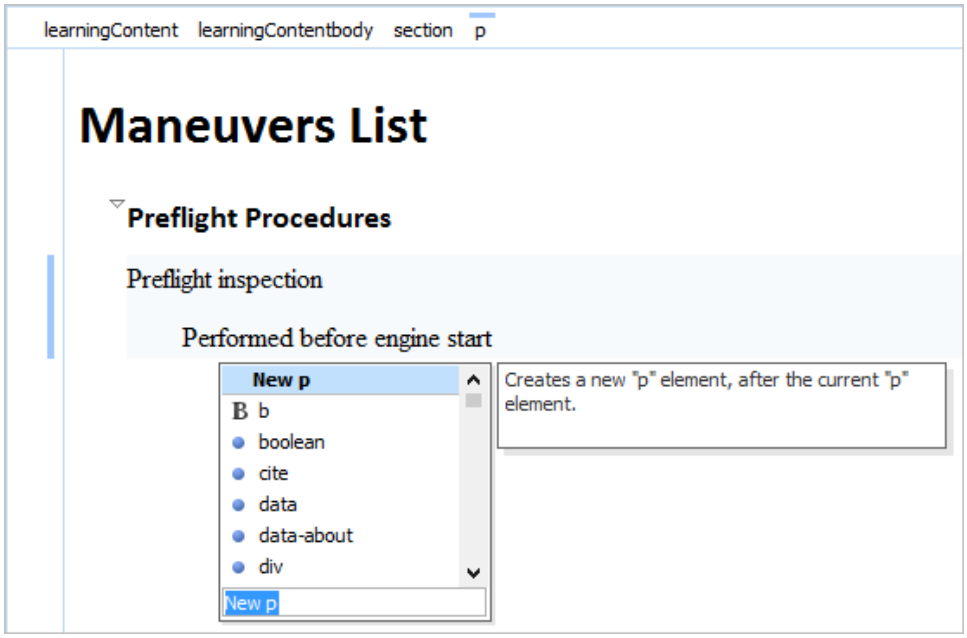
6. At the **sli** placeholder, type a subentry.



7. At the end of the entry, press Enter and then double-click **New sli** in the Content Completion Assistant.

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8. If you want to return to the main entry level, press the Down Arrow key until **p** is displayed at the end of the breadcrumbs, then press Enter and double-click **New p** in the Content Completion Assistant.



9. To create a new maneuvers section, press the Down Arrow key until **learningContentbody** is displayed at the end of the breadcrumbs *and* **section** is available in the **Elements** pane.



10. Repeat steps above as required to complete the maneuvers list.

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23 Creating Procedures (ITM Only)

Note: This section occurs in the ITM only.

The Procedures section is implemented as a series of separate, brief topics and output as one content unit by using nesting and **@chunk** in the map. For more information, see Creating Maps.

Topic nesting may be used to create multiple subsections and hierarchy in the table of contents. The subsections are created by using a single Procedures template.

Three templates are required to complete the Procedures unit:

- Lesson Overview template (for Overview and Objectives).
- Procedures template (for Overlay Topic, Procedure Steps).
- Lesson Debrief template (for Summary).

To create the procedures:

1. From the **Choose Topic** drop-down list, choose the appropriate procedures template listed above.
2. Once created, attach the Procedures file to the appropriate workflow, and then check out the file.

23.1 Title and Navigation Pages

The Procedures title and navigation page, and the sectional title and navigation pages, are implemented in the ITM as automatically generated pages. Example:



23.2 Overview and Objectives

In CACTUS, from the **Choose Topic** drop-down list, choose the **Lesson Overview** template, which is based on the learningOverview DITA topic type. Use the following elements in the template:

<lcIntro>

<lcObjectives>

Delete unused elements.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Overview

1. Inside <lcIntro>, type **Overview** over the **Lesson Summary** heading text.
2. Type the overview text over the **Text** placeholder.
3. Delete any unused paragraph, note, and bullet item placeholders.

Objectives

1. Inside <lcObjectivesGroup><lcObjective>, type the objectives content over the **Objective** placeholder text.

2. If you need to create sub-objectives, type them over the **Sub-objective** placeholder text.

3. To create more objectives list items, on the last item, press Enter, and then double-click **New lcObjective** in the Content Completion Assistant.

Media

To insert a Captivate or Video object, see Inserting Media.

23.3 Creating an Overlay Topic

Many of the topics in Procedure Steps overlay a background image. The procedure for authoring these topics is different from creating ordinary topics.

In CACTUS, from the **Choose Topic** drop-down list, choose the **Procedures** template. Use the elements of the template as required. Copy and paste elements to create additional components. Delete unused elements.

1. On the **CACTUS** menu, choose **Insert Object** > **Insert Media** > **Insert Image and Text Overlay**.

2. In the **Insert Image and Text Overlay** dialog box, select a position for the overlay topic from the **Text overlay position** drop-down list.

3. To specify the background image file, type a file name and then press Enter, or click **Search CACTUS** or **Browse CACTUS**.

4. If you select **Browse CACTUS**, you can browse the directory tree. When you find the image file you want, click **Add**, then go to step 8.

5. If you select **Search CACTUS**, in the **Search Media** dialog box, from the **Media Type** drop-down list, select the **Image** file type.

6. Enter any other search criteria, including **Model**, and then click **Search**.

7. In the list of files found, select a row, or press Ctrl+click to select multiple rows, and then click **Insert**.

8. In the **Insert Media** dialog box, note that the image will be inserted uncropped, with the full size rendering option. Click **Insert**.

   The image is inserted into your document, and a section is created for the overlay topic.

9. Create the topic content in the new section by using the DITA components in your template.

23.4 Procedure Steps

The individual procedure steps are implemented by using a single Procedures template. Create a separate topic per procedure step. Follow the procedure in Creating an Overlay Topic to create topics that are overlaid with positioning on background images.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Headings below the title of the primary components are implemented by using nesting and **@chunk** in the map. For more information, see Creating Maps.

Note: The headings of component sections below the primary component are styled appropriately when the topics are nested in the map and the **@chunk** attribute is used.

References

Insert required document references as described in Inserting References (CAMA and FCOM).

Paragraph

If one or more paragraphs are required, type over the **Text** placeholder.

Instructor Note

Type the note content over the **Instructor note text** placeholder.

In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.

Checklist Table

In the Procedures topic, create the checklist table as a separate section..

1. In <section><title>, type the checklist heading.
2. Delete unused elements.
3. In the table, type the checklist column header text (for example, **Captain** and **First Officer**) over the **Column header** placeholders.

4. Type the table content over the **Table cell** placeholders.

5. To add rows, click in the last row of the table, and then on the **CACTUS** menu, choose **Table** > **Insert Rows**.

6. In the **Number of rows** box, type the number of rows you want to add to the bottom of the table.

7. Make sure the **Position** is set to **Below**, then click **OK**.

Student Note

To insert a student note:

1. Press Enter at the end of a paragraph and then double-click note in the Content Completion Assistant.

2. Following the **Note** label, type the content of the note.

   Use the **@type** attribute to change a standard NOTE to a CAUTION or WARNING. This feature is available for both the ITM ePub and FTM PDF outputs.

   Note: Only the “blank” (for simple NOTE), CAUTION, and WARNING @type selections are implemented in CACTUS.

23.5 Practice

In CACTUS, from the **Choose Topic** drop-down list, choose the **Procedures** template. Use the elements of the template as required. Delete unused elements.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

References

Insert required document references as described in Inserting References (CAMA and FCOM).

Paragraph

Create one or more paragraphs by typing over the **Text** placeholder.

Launch Demo

To insert interactive media linked to a “Launch Demo” icon, see Inserting Media.

23.6 Summary

In CACTUS, from the **Choose Topic** drop-down list, choose the **Lesson Debrief** template. The template is based on the DITA learningSummary topic type. Delete unused elements.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Summary Label

Ignore the **Summary** label; it will not be displayed on output.

Paragraph

Create one or more paragraphs by typing over the **Text** placeholder.

Image

To insert an image, see Inserting Media.

24 Creating Systems Knowledge (ITM Only)

Note: This section occurs in the ITM only.

The Systems Knowledge section is implemented as a series of separate, brief topics and output as one content unit by using nesting and @**chunk** in the map. For more information, see Creating Maps.

Topic nesting may be used to create multiple subsections and hierarchy in the table of contents. The subsections are created by using a single Systems Knowledge template.

Three templates are required to complete the Systems Knowledge unit:

- Lesson Overview template (for Overview and Objectives).
- Systems Knowledge template (for Controls and Indicators, Overlay Topic, Systems Description, and Knowledge Check).
- Assessment template (for Assessment).

To create the systems knowledge:

1. From the **Choose Topic** drop-down list, choose the appropriate systems knowledge template listed above.
2. Once created, attach the Systems Knowledge file to the appropriate workflow, and then check out the file.

Insert required document references as described in Inserting References (CAMA and FCOM).

24.1 Title and Navigation Pages

The Systems Knowledge title and navigation page, and the sectional title and navigation pages, are implemented in the ITM as automatically generated pages. Example:



24.2 Overview and Objectives

In CACTUS, from the **Choose Topic** drop-down list, choose the **Lesson Overview** template, which is based on the learningOverview DITA topic type. Use the following elements in the template:

<lcIntro>

<lcObjectives>

Delete unused elements.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Overview

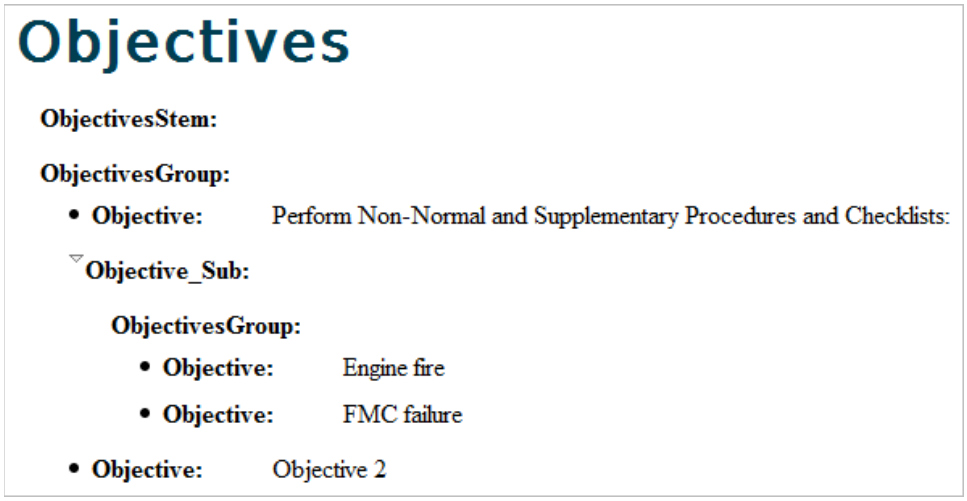
1. Inside <lcIntro><title>, type **Overview** over the **Lesson Summary** heading placeholder text.
2. Type the overview text over the **Text** placeholder.
3. Delete any unused paragraph, note, and bullet item placeholders.

Objectives

1. Inside <lcObjectives><title>, type **Objectives** over the **Lesson Objectives** heading placeholder text

2. If desired, in <lcObjectivesStem>, following the **ObjectivesStem** label, type a clause or sentence introducing the objectives. If an objectives stem is not desired, click the **ObjectivesStem** label, and then press Delete to delete it.

3. Inside <lcObjectivesGroup><lcObjective>, type the objectives content over the **Objective** placeholder text.



4. If you need to create sub-objectives, type them over the **Sub-objectives** placeholder text.

   Note: At this time, only first and second level objectives are available.

5. To create more objectives list items, on the last item, press Enter, and then double-click **New lcObjective** in the Content Completion Assistant.

   To create more sub-objectives, on the last item, press Enter, and then double-click **New lcObjective_Sub** in the Content Completion Assistant.

6. Delete any unused <lcObjective> elements.

24.3 Controls and Indicators

In CACTUS, from the **Choose Topic** drop-down list, choose the **Systems Knowledge** template. Create a separate topic per controls and indicators section. Use the elements of the template as required. Copy and paste elements to create additional components. Delete unused elements.

24.3.1 Creating an Overlay Topic

Many of the topics in the “Controls and Indicators” section overlay a background image. The procedure for authoring these topics is different from creating ordinary topics. For more formation on inserting overlay topics, see Creating an Overlay Topic.

In the segment below the inserted background image, create the overlay topic by using the Systems Knowledge template components, as described in Systems Description.

24.3.2 Systems Description

In CACTUS, from the **Choose Topic** drop-down list, choose the **Systems Knowledge** template. Create a separate topic per systems description section.

The Systems Description section is implemented as a series of separate, brief topics and output as one content unit by using nesting and **@chunk** in the map. For more information, see Creating Maps.

Topic nesting may be used to create multiple subsections and hierarchy in the table of contents. The subsections are created by using a single Systems Knowledge template.

For descriptions of how to use the components of the Systems Knowledge template, see Systems Description.

Title

The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.

Subtitle

For subtitles within major component topics, use <section><title>.

Paragraph

If one or more paragraphs are required, type over the **Text** placeholder.

Instructor Note

If a note to the instructor is required, following the **Note** label, type the note content over the **Instructor note text** placeholder.

If bullets are required under the note, type over the **Instructor note bullet** placeholder text.

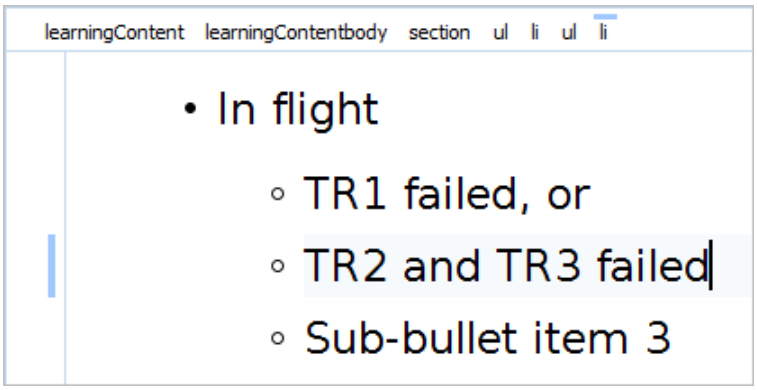
First-Level Bulleted List

Inside , type the list item content over the **Bullet item** placeholder text.

Second-Level Bulleted List

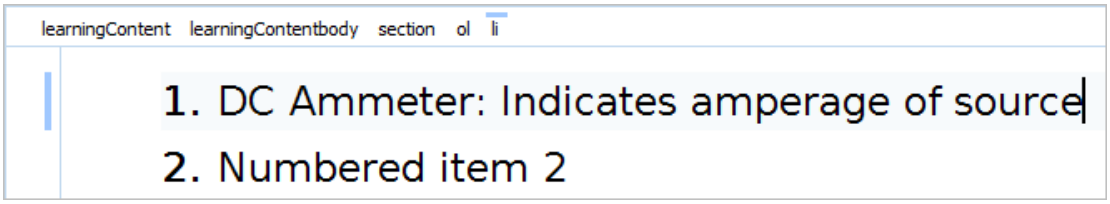
A second-level bulleted list occurs below a first-level bulleted item. The list is contained within the <li> element.

Inside , type the list item content over the **Sub-bullet item** placeholder text.



Numbered List

Inside , type the list item content over the **Numbered item** placeholder text.



Figure

To insert a figure, see Inserting Media.

Student Note

To insert a student note:

1. Press Enter at the end of a paragraph and then double-click note in the Content Completion Assistant.

2. Following the **Note** label, type the content of the note.

   Use the **@type** attribute to change a standard NOTE to a CAUTION or WARNING. This feature is available for both the ITM ePub and FTM PDF outputs.

   Note: Only the “blank” (for simple NOTE), CAUTION, and WARNING @type selections are implemented in CACTUS.

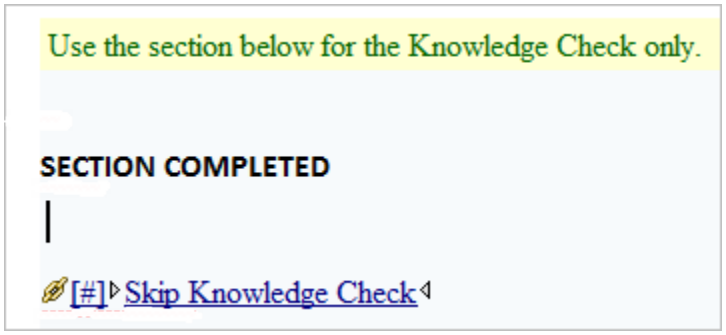
24.4 Knowledge Check

This topic provides a link to a Captivate knowledge check module, and a link to skip the knowledge check and advance to the next topic, if desired.

You create the link to the Captivate object by inserting a media object into a section in the template that is preconfigured for the knowledge check. The title of the section provides a centered heading, and a paragraph element allows you to create a cross-reference to skip the knowledge check and continue to another topic.

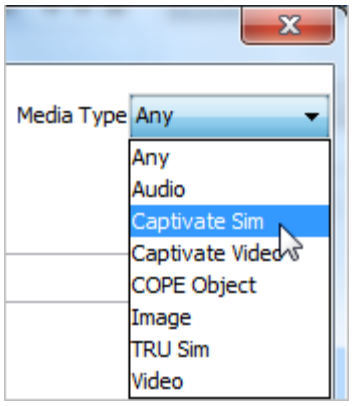
The Knowledge Check section is the last section in the Systems Knowledge template. To add the knowledge check:

1. Check out the appropriate Systems Knowledge file.
2. At the bottom of the file, click below the **SECTION COMPLETED** heading:



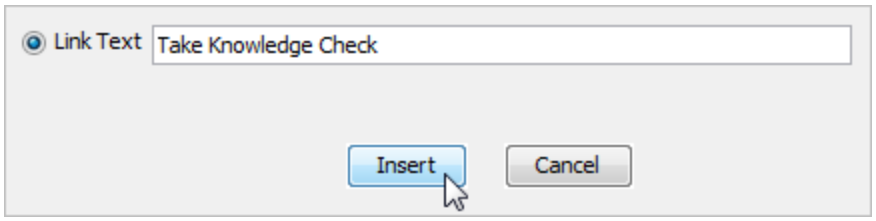
3. Insert the Captivate object. For information on inserting media, see Inserting Media.

   Note: If you choose to search for the media object, on the **Media Type** drop-down list in the upper right corner of the **Search Media** dialog box, select **Captivate Sim**.



4. In the Insert Media dialog box, click Link Text.
5. In the **Link Text** box, type **Take Knowledge Check**, and then click **Insert**.

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6. To link the skip paragraph to the next section, in <section_knowledgecheck><p>, select Skip Knowledge Check.

7. Insert a cross-reference to the section file that you want to link to if the user chooses to skip the knowledge check. For information on inserting the cross-reference, see Inserting Cross-Reference Links.

24.5 Assessment

The Assessment is implemented by using the Assessment template. The template is based on the learningAssessment topic type.

The template contains standard text content, which occurs unchanged in all Assessment topics. In two locations in the template, you type the current chapter name over placeholder text. You also provide a link to the external Learning Management System (LMS).

To create the assessment:

1. From the **Choose Topic** drop-down list, choose the **Assessment** template.

2. Once created, attach the Assessment file to the appropriate workflow, and then check out the file.

3. In the first two paragraphs of standard text, select the **[Chapter Name]** placeholder (including the brackets) and type the current chapter name, replacing both the text and the brackets.

4. In the second paragraph, select the current chapter name that you typed.

5. Add an **@href** attribute with the value of the target URL of the LMS. For information on adding attribute values, see Adding Attributes.

Note: The feature for linking to the assessment is in work; do not use at this time.

25 Creating a Glossary

The Glossary template is used to create individual glossary entries, not a file containing all entries. Create one glossary topic file per glossary term. The glossary topic files are compiled by references in a separate glossary map.

1. From the **Choose Topic** drop-down list, choose the **Glossary** template.

2. Once created, attach the Glossary file to the appropriate workflow, and then check out the file.

Title

The Glossary template does not contain a separate <title> element. The term (<glossterm> element) is displayed as the topic title.

To create the Acronym List, first create a topic with only a <title> of **Acronym List** and no other content. This will become the <topicref> container for the rest of the glossary topics.

Term

In <glossentry><glossterm>, type the term to be defined over the **APU** placeholder text.

Definition

In <glossentry><glossdef>, type the definition over the placeholder definition.

Spelled-Out Form

The <glossbody><glossSurfaceForm> element is provided to allow you to display the spelled-out form of an acronym, followed by the acronym, the first time that the term occurs in a topic. After the first occurrence, the acronym only is shown.

1. At the **Surface form** label, type the spelled-out form of an acronym, followed by the acronym in parentheses, over the placeholder text.

   Note: The **Surface form** label is a guide only; it will not be displayed in output.

2. If the term is not an acronym, use the breadcrumb to select **glossbody**, and then press Delete to delete both <glossbody> and <glossSurfaceForm>.

26 Creating Maps

Map files are initiated in CACTUS. For the procedures for initiating files and opening them to edit in oXygen, see Managing Files and Folders.

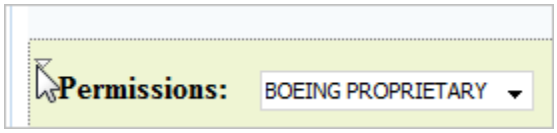
When you open a map or book map template in CACTUS, the CACTUS menu in oXygen reflects the map context.

26.1 Creating a Map

Maps are used to collect topics into a chapter. A book map is used to collect all chapters into a publication. For information on book maps, see Creating a Book Map.

To create a chapter map:

1. Click the **Create Map** button on the menu bar.
2. Once created, attach the map file to the appropriate workflow, and then check out the file.
3. Collapse the block of map metadata:



4. Click at the end of the **Meta information** label:



5. On the **CACTUS** menu, choose **Insert Object** > **Insert Map** or **Insert Topic**.

   Note: Inserting a map into a map is valid, but TD only inserts topics into maps for the FTM and ITM.

6. In the **Insert Topic/Map** dialog box, you can type or paste the file name into the text box, click **Browse Tree UI** to navigate the CACTUS repository to the file that you want to insert, **or Search** to search for the file.
7. If you select **Browse Tree UI**, you can browse the directory tree. When you find the file you want, click **Add**, then go to step 10.
8. If you select **Search**, in the **Search Topic/Map** dialog box, enter all other search criteria, and then click **Search**.
9. In the list of files found, select a row, or press Ctrl+click to select multiple rows, and then click **Insert**.

Your selection is added to the **Insert Topic/Row** dialog box.

10. In the **Insert Topic/Row** dialog box, you can

    - Select an item and then click **Remove** to remove it from the list.
    - Click **Clear** to clear the entire list.
    - Select an item and then click the up/down arrow buttons to reorder the list. The topics or maps will be inserted into your document in the order shown in the dialog box.

11. When you are ready to insert the items into your document, click **Insert All**.

12. Repeat the previous steps to add all desired topics or maps.

    Note: You build a glossary map in the same way, by inserting topic references to the individual glossary entry topics in alphabetical order. For more information on glossaries, see Creating a Glossary.

    Note: See also Nesting Topic References to learn how to create different levels of content.

    Note: See also Chunking Into Output Units to learn how to group topics together into a single unit of output like a page or a longer section.

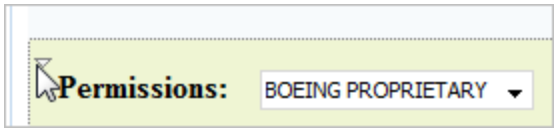
13. Save the map in CACTUS.

26.2 Creating a Book Map

Book maps combine chapter maps into a publication. Part and chapter titles that you specify in the book map are displayed in the navigation. Together, the book map and their underlying maps provide a full hierarchy of navigation links.

To create a book map:

1. Click the **Create Book Map** button on the menu bar.
2. Once created, have your TD Lead create and forward the Editing or Express workflow.
3. Acquire and open the appropriate workflow, and then check out the file.
4. Collapse the block of book map metadata:



5. Click at the end of the **Meta information** label:



6. Press Enter.

   The Content Completion Assistant is displayed.

7. Double-click **chapter** (or **part**, if a volume-level element is required).

   A new chapter element and **@id** are inserted.

   Maps and topic titles are used automatically as titles in the book map. Part and chapter elements do not exist in the CACTUS repository, however. You *must* enter the titles of those components manually.

8. Click at end of the chapter (or part) and add a **topicmeta** element. For more information on adding elements, see Inserting Elements.

9. With the insertion point in the **topicmeta** element, add a **navtitle** element.

10. At the **Navigation Title** label, type the title text of the chapter or part.

    This title text will be used in the navigation.

11. If you are creating an ITM book map, chapters and parts are preceded by divider pages with background images. Specifically for chapters, you can select the background image by using a CACTUS menu command.

    a. With the insertion point still in the **topicmeta** element, on the **CACTUS** menu, choose **Insert Object** > **Insert Title Image**.

    b. In the **Insert Media** dialog box, browse or search for the background image, and then click **Insert**.

12. If you have inserted a part, click at the end of the part element and then perform steps 6 through 11b to insert a **chapter** element, title, and background image.

13. To insert a chapter map, click at the end of the chapter line, and then on the **CACTUS** menu, choose **Insert Object** > **Insert Map**.

14. To select and insert the map file, follow steps 5 through 11 in Creating a Map. Select only chapter map files, not topic files.

15. Repeat the previous steps to add all desired chapter maps.

16. Save the book map in CACTUS.

26.3 Nesting Topic References

You can create hierarchy in your navigation by nesting <topicref>s in the map. Nested topicrefs will be displayed at corresponding levels of indent in the navigation.

To nest a topic reference:

1. Click inside the higher level (container) topicref, and then on the **CACTUS** menu, choose **Insert Object** > **Insert Topic** to locate and insert a reference to the subtopic file.

2. Select and insert the topic file by following steps 5 through 11 in Creating a Map .

   The topicref is inserted nested in the container topicref.

   Note: Though oXygen will allow you to nest multiple layers of topics, only one level of nested topics is valid for publishing.

To insert a topic reference that is not nested:

1. Click at the end of a topicref, and then press the Right Arrow or Down Arrow key to move the insertion point to the next line.

2. On the **CACTUS** menu, choose **Insert Object** > **Insert Topic** to locate and insert a reference to a topic file.

3. Select and insert the topic file by following steps 5 through 11 in Creating a Map.

   The topicref will be inserted parallel to (as a sibling of) the preceding topicref.

26.4 Chunking Into Output Units

Nested topics can be combined into a single content unit.

Note: This is required for the ITM and FTM manuals.

1. Create a container topic with nested topics, as described in Nesting Topic References.

2. On the container topicref, add a **@chunk** attribute with a value of **to-content**. For more information on adding attributes, see Adding Attributes.

   On output, the nested topics will be combined into a single topic. Only the title of the container topic will be displayed in the navigation.

27 Publishing Training Content

After you have checked in all files to the CACTUS repository, you can publish them by using the CACTUS Assemble function.

See Performing Assemblies for details on using Assemble to publish training content.

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Appendix A Training Development Authoring Quick Reference

A.1 CACTUS Menus

This segment provides a reference to the commands on the CACTUS menu in oXygen Author mode. Many of the commands are available on right-click pop-up menus also, as well as through the CACTUS toolbar. The menu is not displayed in Text mode.

The CACTUS menu responds to the current information type. Some of the commands are different, depending on whether a topic or a map/book map is being edited.

A.1.1 Topic Menu

Table 27-1 CACTUS Menu Commands: Topics

Command	Description
 Insert Object	Command group.
Insert Media	Allows you to search or browse the CACTUS repository to insert media files, including images, videos, and Captivate Sim objects.
 Insert Element	Command group.
Insert Paragraph (Ctrl+Shift+P)	Inserts a paragraph element.*
Insert Section	Inserts a section element.*
Insert Equation	*Feature in work; do not use at this time.*
Insert List Item	Inserts a list element inside an ordered or unordered list.*
Insert Unordered List	Inserts an unnumbered bulleted list.*
Insert Ordered List	Inserts a numbered list.*
Insert Glossary Term	Allows you to search or browse to insert an existing glossary term, or create a new glossary term and definition.
 Configure	Command group.
Insert Revision Marking	Applies revision marking to an added, modified, or deleted element.

* This method of inserting the element will create a valid DITA context automatically if the location of the insertion is invalid. Note that the result may be unexpected. Alternative methods may be preferable: selecting an element from the Elements view pane, or pressing Enter and selecting an element from the Content Completion Assistant.

Command	Description
Detach Revision Marking	Removes revision marking that has been applied to an element.
Attach Applicability (Ctrl+Shift+A)	Allows you to search or browse the CACTUS repository to insert an existing applicability Boolean expression, or create a new Boolean from configuration items.
Detach Applicability (Ctrl+Shift+D)	Removes applicability that has been applied to an element.
 Views	Command group.
Print Preview	*Feature in work; do not use at this time.*
BSEC Check	*Feature in work; do not use at this time.*
Current Environment	Identifies the current CACTUS environment (dev, prod, etc.).
 Link	Command group.
Cross-Reference	Allows you to link a selection to an element in the current topic or in another topic in the CACTUS repository.
 List	*Feature in work; do not use at this time.*
 Table	Command group.
Insert Table	Inserts a CALS-style or simple table with the title, number of rows, and table formatting properties that you specify.
Insert Row Above	Inserts a row above the current row.
Insert Row Below	Inserts a row below the current row.
Insert Rows	Inserts the number of rows that you specify either above or below the current row.
Delete Row(s)	Deletes one or more rows; the current row, and any columns that contain selected cells (<entry> elements) will be deleted.
Insert Column Before	Inserts a column to the left of the current column.
Insert Column After	Inserts a column to the right of the current column.
Insert Column(s)	Inserts the number of columns that you specify either to the left or the right of the current column.
Delete Column(s)	Deletes one or more columns; the current column, and any columns that contain selected cells (<entry> elements) will be deleted.
Insert Cell	*This standard oXygen feature is not useful in most circumstances. Inserting a row or column should provide the results you need*.
Join	Merges two or more adjacent selected cells in rows and columns; at least two adjacent cells must be selected.
Split	Allows you to split the current cell or joined cells. If the selection can be split in more than one way, a dialog

Command	Description
	box allows you to specify the number of resulting rows and columns.
Table Properties (Ctrl+T)	Allows you to view and change table display properties such as alignment in cells, row and column separators, and table frame. In general, settings of child elements take precedence; for example, settings of a cell in a row take precedence over settings of the row.
ID Options	*Feature in work; do not use at this time.*
Browse DITA Style Guide	Links to *The DITA Style Guide Best Practices for Authors*, by Tony Self.
 Refresh References	Refreshes cross-references and key references to objects in the CACTUS repository.
 Tags Display Mode	Command group.
Full Tags with Attributes	Displays full tag names with attributes for both block level and in-line level elements.
Full Tags	Displays full tag names without attributes for both block level and in-line level elements.
Block Tags	Displays full tag names for block level elements and simple tags without names for in-line level elements.
Inline Tags	Displays full tag names for inline level elements, while block level elements are not displayed.
Partial Tags	Displays simple tags without names for in-line level elements, while block level elements are not displayed. This is the default mode for Training Development.
No Tags	No tags are displayed. This is the most compact mode and closest to a word processor view.
 Profiling/Conditional Text	*Feature in work; do not use at this time.*

A.1.2 Map and Book Map Menu

Table 27-2 CACTUS Menu Commands: Maps and Book Maps

Command	Description
 Insert Object	Command group.
Insert Map	Allows you to search or browse the CACTUS repository to insert a DITA map.
Insert Topic	Allows you to search or browse the CACTUS repository to insert a DITA topic.
 Configure	Command group.
Insert Revision Marking	Applies revision marking to an added, modified, or deleted element.
Detach Revision Marking	Removes revision marking that has been applied to an element.
Attach Applicability (Ctrl+Shift+A)	Allows you to search or browse the CACTUS repository to insert an existing applicability Boolean expression, or create a new Boolean from configuration items.
Detach Applicability (Ctrl+Shift+D)	Removes applicability that has been applied to an element.
Insert	*Feature in work; do not use at this time.*
 Relationship Table	*Feature in work; do not use at this time.*
 Refresh References	Refreshes cross-references and key references to objects in the CACTUS repository.
 Tags Display Mode	Command group.
Full Tags with Attributes	Displays full tag names with attributes for both block level and in-line level elements.
Full Tags	Displays full tag names without attributes for both block level and in-line level elements.
Block Tags	Displays full tag names for block level elements and simple tags without names for in-line level elements.
Inline Tags	Displays full tag names for inline level elements, while block level elements are not displayed.
Partial Tags	Displays simple tags without names for in-line level elements, while block level elements are not displayed. This is the default mode for Training Development.
No Tags	No tags are displayed. This is the most compact mode and closest to a word processor view.
 Profiling/Conditional Text	*Feature in work; do not use at this time.*

A.2 Special Characters

The following special characters should not be used in the property fields:

- angle brackets (< >)
- ampersand (&)
- em-dash (—)
- en-dash (–)
- degree symbol (◦)
- backslash (\)
- forward slash (/)
- asterisk (*)
- question mark (?)

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A.3 Common Procedures

A.3.1 Inserting Elements

Method	Action
Use Elements view pane	Click in topic where you want to add element. In **Elements** view pane, double-click element name.
Use Content Completion Assistant	Click in topic where you want to add element (for example, at end of an existing element), then press Enter. In Content Completion Assistant, use combination of text entry and Up and Down Arrow keys to select element, then press Enter.

A.3.2 Selecting an Element by Using the Breadcrumb

Action
At top of authoring window below file name tab, click node of breadcrumb to select element.

A.3.3 Adding Attributes

Method	Action
Use Attributes view pane	Select element to which you want to add attribute. In **Attributes** view pane (upper right corner of oXygen window), double-click in **Value** cell to right of attribute name. Type value for attribute, or select value from drop-down list.
Use in-place attributes editor	Select element to which you want to add attribute. Press Alt+Enter, or right-click element and choose **Edit Attributes** from pop-up menu. In in-place attributes editor, in **Name** box, select attribute from drop-down list. In **Value** box, type value for attribute, or select value from drop-down list. Click **OK**.

A.3.4 Inserting References (CAMA and FCOM)

Action
Click in element where you want to add reference. Depending on document type to reference, add either **@camaId** or **@fcomId** attribute. See Adding Attributes. In attribute **Value** box, type CAMA or FCOM ID value, then press Enter. *To insert more than one CAMA or FCOM value, separate values with a comma.*

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A.3.5 Inserting Cross-Reference Links

Action
Select text that you want to link. Choose **CACTUS** > **Link** > **Cross-Reference**. In **Cross reference (xref)** dialog box, browse to external target file in repository or select target element in current document. If you select external file in repository, list of target elements updates to show elements in external file. Select target element in list, then click **OK**. *To remove cross-reference, click at beginning of element where cross-reference was inserted, then press Backspace.*

A.3.6 Validating Links and Content

Action
oXygen flags invalid or broken hyperlinks with yellow or red horizontal bars in scroll bar at location of broken links or other invalid content. Click on bar. oXygen scrolls to location and highlights it. Cycle through bars by clicking blue navigation arrows at bottom of scroll bar. To perform validation of all topic content on demand, click **Validate** button on toolbar or press Ctrl + Shift + V.

A.3.7 Attaching and Removing Applicability

Action
Click in element where you want to add applicability or just above it. Choose **CACTUS** > **Configure** > **Attach Applicability**, or press Ctrl+Shift+A. In **Boolean selection builder** dialog box, select **Search Booleans** or **Search CIs**. Enter search term (for example, **reg**) in text box and then click **Search**. In list of search results, select item then click **Insert CI or Boolean**, or double-click item. If you have selected CI to build new Boolean expression, below list, click logical operator button to establish relation to next expression or CI. Click **Validate** at any time to check whether expression has been properly constructed. When you have selected or built complete Boolean, click **Attach.** *To remove applicability that you have attached to element, click in element, then choose **CACTUS** > **Configure** > **Detach Applicability**, or press Ctrl+Shift+D. In dialog box, click **Yes**.*

A.3.8 Using Revision Marking

Action
Adding Revision Marking Click on line where you want to add revision marking. Choose **CACTUS** > **Configure** > **Insert Revision Marking**. In **Reason for Revision** dialog box, select whether revision marking is **Visible** or **Hidden**. Select revision type: **Added**, **Modified**, or **Deleted**. In **Reason for Revision** text box, enter reason for revision. Click **Save**. Modifying Revision Marking Click on line where you added revision marking. Choose **CACTUS** > **Configure** > **Insert Revision Marking**. Modify any existing selections, then click **Save.** Removing Revision Marking Click on line where you added revision marking. Choose **CACTUS** > **Configure** > **Detach Revision Marking**. In **Confirmation** dialog box, click **Yes**.

A.3.9 Inserting Media

Action
Click where you want to insert media object. Choose **CACTUS** > **Insert Object** > **Insert Media**. In **Insert Media** dialog box, type file name then press Enter, or click **Search CACTUS** or **Browse CACTUS**. When you have selected media object file, click **Add** or **Insert**. In **Insert Media** dialog box, specify cropping and rendering options. Click **Insert**.

A.3.10 Creating Image Hotspots (ITM Only)

Action
Insert image on which you want to create hotspot link. See Inserting Media. Right-click image, then choose **Image Map Editor** from pop-up menu. In **Image Map Editor** dialog box, select shape button: **New Rectangle**, **New Circle**, or **New Polygon**. Move shape to desired area in image and drag points on shape to resize and adjust it. With shape selected, on **Link** menu at right end of **Target** box, choose **File Reference** linking option. In **File Reference** dialog box, click folder button at right end of **Location** box. At bottom of drop-down menu, click **CACTUS**. Browse directory structure to locate file that you want to link to from hotspot. Select file, then click **Add**. In **File Reference** dialog box, click **OK**. To add more hyperlinks, select shape button again and repeat appropriate steps. When finished creating hyperlinks, click **OK**.

A.3.11 Styling Inline Text

Action
Creating Indications Select text to which you want to apply indication styling. In **Elements** view pane, double-click **ph**. Add @**indications** attribute to **ph** element. See Adding Attributes. In **Value** box, select indication type from drop-down list. Click **OK** to close attributes editor. Applying Subscript and Superscript Select text to which you want to apply subscript or superscript. In **Elements** view pane, double-click **sub** to apply subscript or **sup** to apply superscript.

A.3.12 Specifying a Document Type

Action
Click in element that you want to make specific to document type. Add @**deliveryTarget** attribute with value of either **ITM-ePub** or **FTM-PDF**. See Adding Attributes.

A.3.13 Syllabus (FTM)

CACTUS: **Choose Topic** > **Syllabus** template.

Component	Action
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
Day Lists	**Headings** DAY 1 heading pre-entered in <section><title>. Classroom subheading pre-entered in <section><p>. **Syllabus List** Type over **Item** to create Syllabus list. To add new items, at end of entry, press Enter, then double-click **New sli**. Copy, paste, and revise elements to build Syllabus list. Delete unused sections and placeholder text.

A.3.14 Syllabus (ITM)

CACTUS: **Choose Topic** > **Syllabus** template.

Component	Action
Topic	Delete section at top of template, provided for FTM Syllabus.
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
List of Day Headings	Generated from day list section headings that follow.

Component	Action
Day Lists	**Headings** In each of section (<section_introduction>, e.g.), type heading in <title>. **Syllabus List** Type over Module and Profile placeholder **Text** to create Syllabus list entries. To add new items, at end of entry, press Enter, then double-click **New p**. Delete unused sections and placeholder text.

A.3.15 Introduction

Use these instructions if you create a new Introduction file.

CACTUS: **Choose Topic > Learning Content** basic topic type.

Component	Action
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
Section	Click at **learningContentbody**. In **Elements** pane, double-click **section**. Type over **title** placeholder to create sectional heading.
Paragraph	Type over **p** placeholder. To add new paragraph, at end of paragraph, press Enter, then double-click **New p**.
Bulleted List	At end of paragraph, press Enter, then double-click **ul**. At **li** placeholder, type list item content. To add new list item, at end of item, press Enter, then double-click **New li**.
Student Note	At end of paragraph or list item, press Enter, then double-click **note**. Following **Note** label, type note content. Use the **@type** attribute to change a standard NOTE to a CAUTION or WARNING. This feature is available for both the ITM ePub and FTM PDF outputs. **Note:** Only the “blank” (for simple NOTE), CAUTION, and WARNING @type selections are implemented in CACTUS. *Student notes are visible to both students and instructors, unless value of **@audience** attribute is set to “Student.” In this rare case, note is visible to students only*.

A.3.16 Lesson Overview

CACTUS: **Choose Topic** > **Lesson Overview** template.

Component	Action
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
Lesson Summary	<lcIntro> element. (“Intro:” is not displayed in output.) **Heading** Pre-entered. **Paragraph** In <lcIntro><p>, type over **Text**. **Instructor Note** In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience=“Instructor”**. Once tagged with **@audience=“Instructor”**, this content will be shown in outputs for instructors but hidden in non-instructor outputs. **Bulleted List** Type over **Bullet item**.
Duration	<lcDuration> element. (“Duration:” is not displayed in output.) **Heading** Pre-entered. **lcTime** Enter hours and minutes. End each duration with period.
Prerequisites	<lcPrereqs> element. (“Prereqs:” is not displayed in output.) **Heading** Pre-entered. **Prerequisites List** Type over **Prerequisite**.
Lesson Objectives	<lcObjectivesGroup><lcObjective> element. (“Objectives:” is not displayed in output.) **Heading** Pre-entered. **ObjectivesGroup** Following each **Objective** label, type learning objective. To add new objectives, at end of entry, press Enter, then double-click **New lcObjective**. Delete unused <lcObjective> elements.
Assessment and Minimum Proficiency	<section> element. **Heading** Pre-entered. **Paragraph** In <section><p>, type over **Text**.

A.3.17 Lesson Data

CACTUS: **Choose Topic** > **Lesson Data** template.

Component	Action
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
Route of Flight	**Heading** Pre-entered. Type route in parentheses; example: **(KSEA-KMWH)**. **Route Table** Enter values in right column in <entry> element. Delete 717 Flight Plan table if not used.
Performance	**Heading** Pre-entered. **Performance Table** Use table specific to model. Enter values in right column in <entry> element. Delete unused tables.
Weather	**Heading** Pre-entered. **Weather Table** For 737CL, 747, 757, 767, 777, and 787, enter labels in left column. Tables with standard labels in left column have been provided for 737NG, 737 MAX, and 717. Enter values in right column in <entry> element. **Instructor Note** In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs. **Paragraph** Type over **Text** or delete **p** if no paragraph required. Delete unused tables.
Clearance	**Heading** Pre-entered. **Paragraph** In <section><p>, type over **Text** placeholder.

A.3.18 Modules (ITM Only)

CACTUS: **Choose Topic** > **Modules** template.

Component	Action
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
Headings	Pre-entered. Styling and color are applied on output based on “module_*abc*” heading element used.
Bulleted List	Type over **Bullet item** and **Sub-bullet item.** To add new list item, at end of item, press Enter, then double-click **New li**. Delete unused list items.

A.3.19 Lesson Plan

CACTUS: **Choose Topic** > **Lesson Plan** template.

Component	Action
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
Flight Profile Images	One topic per flight profile image. **Title** The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. **Figure** In oXygen, choose **CACTUS** > **Insert Object** > **Insert Media**.
Track Image	One topic per track image. **Title** The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. **Figure** In oXygen, choose **CACTUS** > **Insert Object** > **Insert Media**.

Component	Action
Track Data	All phases of flight in separate sections in single topic. **Title** The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. **Phases of Flight** In <section><title>, type phase of flight heading (**Preflight**, for example) over **title**. Type phase of flight entries over **Numbered item.** Manually numbered. If instructor notes are required, in addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, @**audience="Instructor"**. Once tagged with @**audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.
Device Familiarization	One topic per device familiarization section. **Title** The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. **Device Topics** In <section><title>, type phase of flight heading (**Flat Panel Trainer**, for example) over **title**. Type device topics over **Bullet item.** If instructor notes are required, in addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, @**audience="Instructor"**. Once tagged with @**audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.

Component	Action
Flight Profile Segment Sections	One topic per flight profile segment. **Title** The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. **Step** In <section><p> type step content over **Step Text**. Number steps sequentially across subsections. **Bulleted List** Type over **Bullet item**. To add new list item, at end of item, press Enter, then double-click **New li**. **Instructor note** In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs. **Student note** Use the **@type** attribute to change a standard NOTE to a CAUTION or WARNING. This feature is available for both the ITM ePub and FTM PDF outputs. **Note:** Only the "blank" (for simple NOTE), CAUTION, and WARNING @type selections are implemented in CACTUS.

A.3.20 Lesson Debrief

CACTUS: **Choose Topic** > **Lesson Debrief** template.

Component	Action
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
Summary Label	Ignore; not displayed in output.
Headings	Standard headings (Student Evaluation of Session, and so on) have been pre-entered.
First-Level Bulleted List	In , type over **Bullet item**.
Instructor Note	In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs.
Second-Level Bulleted List	These questions occur below notes. In <lcInstructornote2> , type over **Question**.
Third-Level Bulleted List	These questions occur below bulleted questions. In <lcInstructornote2> , type over **Subquestion**.

A.3.21 Evaluation

Four templates are required to complete the Evaluation unit:

- CACTUS: **Choose Topic** > **Lesson Overview** template (for Evaluation Overview).
- CACTUS: **Choose Topic** > **Lesson Data** template (for Evaluation Data).
- CACTUS: **Choose Topic** > **Lesson Plan** template (for Evaluation Profile).
- CACTUS: **Choose Topic** > **Learning Content** basic topic type (for Maneuvers List).

Component	Action
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
Lesson Overview template	
Evaluation Overview	Follow procedures in Lesson Overview.
Lesson Data template	
Evaluation Data	Follow procedures in Lesson Data.
Lesson Plan template	
Evaluation Profile	Follow procedures in Flight Profile Images.
Learning Content basic topic type	
Maneuvers List	**Title** The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. **List of Maneuvers** Click at **learningContentbody**. In **Elements** pane, double-click **section**. Type over **title** placeholder to create maneuver section heading; for example, **Preflight Procedures**. Type over **p** placeholder to create main entry in maneuvers list. To create subentries, click at end of main entry, and in **Elements** pane, double-click **sl**. At **sli** placeholder, type subentry. To add new subentry, at end of item, press Enter, then double-click **New sli**. To return to main entry level, press Down Arrow key until **p** is displayed at end of breadcrumbs, then press Enter and double-click **New p**. To create new maneuvers section, press Down Arrow key until **learningContentbody** is displayed at end of breadcrumbs and **section** is available in **Elements** pane. Double-click **section**.

A.3.22 Procedures (ITM Only)

Three templates are required to complete the Procedures unit:

- CACTUS: **Choose Topic > Lesson Overview** template (for Overview and Objectives).
- CACTUS: **Choose Topic > Procedures** template (for Overlay Topic, Procedure Steps, and Practice).
- CACTUS: **Choose Topic > Lesson Debrief** template (for Summary).

Component	Action
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
Lesson Overview template	
Overview and Objectives	<lcIntro> element. **Overview Heading** Type **Overview** over **Lesson Summary** heading text. **Paragraph** Type overview text over **Text**. Delete unused paragraph, note, and bullet item placeholders. <lcObjectives> element: **Objectives Heading** Pre-entered. Delete "Lesson." **Objectives and Sub-Objectives** Type over **Objective**. Type over **Sub-objective**. To add new objectives, at end of last objective, press Enter, then double-click **New lcObjective**. **Media** To insert Captivate object, choose **CACTUS** > **Insert Object** > **Insert Media**.
Procedures template	
Overlay Topic	Choose **CACTUS** > **Insert Object** > **Insert Media** > **Insert Image and Text Overlay**. Select position for overlay topic from **Text overlay position** drop-down list. To specify background image file, type file name, search, or browse repository. To insert image (full size rendering option), click **Insert**. Image is inserted, and section is created for overlay topic. Create topic content in new section by using DITA components in template.

Procedures, continued

Component	Action
Procedures template	
Procedure Steps	One topic per procedure step. **Title** The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. **References** Click in element where you want to add CAMA or FCOM reference. Depending on document type, add either **@camaId** or **@fcomId** attribute. In attribute **Value** box, type CAMA or FCOM ID value, then press Enter. **Paragraph** Type over **Text** placeholder. **Instructor Note** In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs. **Student note** Use the **@type** attribute to change a standard NOTE to a CAUTION or WARNING. This feature is available for both the ITM ePub and FTM PDF outputs. **Note:** Only the "blank" (for simple NOTE), CAUTION, and WARNING @type selections are implemented in CACTUS. **Checklist Table** Create checklist table as separate topic. The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. In <section><title>, type checklist heading. Delete unused elements above table. In table, type checklist column header text (e.g., **Captain**) over **Column header** placeholders. Type table content over **Table cell** placeholders. To add rows, click in last row of table then choose **CACTUS** > **Table** > **Insert Rows**. In **Number of rows** box, type number of rows you want to add. With **Position** set to **Below**, click **OK**.

Procedures, continued

Component	Action
Practice	**Title** The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. **References** Click in element where you want to add CAMA or FCOM reference. Depending on document type, add either **@camaId** or **@fcomId** attribute. In attribute **Value** box, type CAMA or FCOM ID, then press Enter. **Paragraph** Type over **Text** placeholder. **Launch Demo** Choose **CACTUS** > **Insert Object** > **Insert Media**. Insert interactive media linked to “Launch Demo” icon.
Lesson Debrief template	
Summary	**Title** The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS. **Summary Label** Ignore; not displayed on output. **Paragraph** Type over **Text** placeholder. **Figure** In oXygen, choose **CACTUS** > **Insert Object** > **Insert Media**.

A.3.23 Systems Knowledge (ITM Only)

Three templates are required to complete the Systems Knowledge unit:

- CACTUS: **Choose Topic > Lesson Overview** template (for Overview and Objectives).
- CACTUS: **Choose Topic > Systems Knowledge** template (for Controls and Indicators, Overlay Topic, Systems Description, and Knowledge Check).
- CACTUS: **Choose Topic > Assessment** template (for Assessment).

Component	Action
Title	The topic title is created automatically from the Title that you enter when you begin a topic in CACTUS.
Lesson Overview template	
Overview and Objectives	<lcIntro> element: **Overview Heading** Type **Overview** over **Lesson Summary** heading text. **Paragraph** Type overview text over **Text**. Delete unused paragraph, note, and bullet item placeholders. <lcObjectives> element: **Objectives Heading** Pre-entered. Delete “Lesson.” **Objectives and Sub-Objectives** Type over **Objective**. Type over **Sub-objective**. To add new objectives, at end of last objective, press Enter, then double-click **New lcObjective**.
Systems Knowledge template	
Controls and Indicators	One topic per controls and indications section. Use template elements as required. Copy and paste elements to create additional components. Delete unused elements
Overlay Topic	Follow procedures in Overlay Topic.

Systems Knowledge, continued

Component	Action
Systems Knowledge template	
Systems Description	**Subtitle** Use <section><title>. **Paragraph** Type over **Text**. **Instructor Note** In addition to using the <lcInstructornote2> element, Instructor Notes must be tagged with the attribute, **@audience="Instructor"**. Once tagged with **@audience="Instructor"**, this content will be shown in outputs for instructors but hidden in non-instructor outputs. **Student note** Use the **@type** attribute to change a standard NOTE to a CAUTION or WARNING. This feature is available for both the ITM ePub and FTM PDF outputs. **Note:** Only the "blank" (for simple NOTE), CAUTION, and WARNING @type selections are implemented in CACTUS. **First-Level Bulleted List** In , type over **Bullet item**. **Second-Level Bulleted List** This list occurs below first-level bulleted item. In , type list item content over **Sub-bullet item**. **Numbered List** In , type over **Numbered item**. **Figure** In oXygen, choose **CACTUS** > **Insert Object** > **Insert Media**.
Knowledge Check	**Title** Title is generated. **Captivate Object** Click below **SECTION COMPLETED** heading. Choose **CACTUS** > **Insert Object** > **Insert Media**. Insert **Captivate Sim** with **Text Link** option. As link text, type **Take Knowledge Check**, then click **Insert**. **Skip Knowledge Check** In <section_knowledgecheck><p>, select **Skip Knowledge Check**. Choose **CACTUS** > **Link** > **Cross-Reference** to insert link to next logical section following Knowledge Check.

Assessment template	
Assessment	In first two paragraphs of standard text, type current chapter name over **[Chapter Name]** placeholder. In second paragraph, select current chapter name and add **@href** attribute with value of target URL of LMS. *The feature for linking to the assessment is in work; do not use at this time.*

A.3.24 Glossary

CACTUS: **Choose Topic** > **Glossary** template.

Component	Action
Title	One glossary topic per glossary term. Term (<glossterm> element) is displayed as topic title.
Term	In <glossentry><glossterm>, type term to be defined over **APU** placeholder.
Definition	In <glossentry><glossdef>, type definition over placeholder definition.
Spelled-Out Form	At **Surface form** label, type spelled-out form of an acronym, followed by acronym in parentheses, over placeholder (optional). If term is not an acronym, select **glossbody**, then press Delete to delete both <glossbody> and <glossSurfaceForm>.

Appendix B DITA Methods and Best Practices

The following segments describe DITA features and authoring techniques that support single sourcing and reuse.

B.1 Titling Topics

A title is required for all topics. Typically, the title is rendered as the heading of the content and is displayed in the table of contents and other navigation elements. The title must clearly convey the content of the topic.

When writing titles, follow these guidelines:

- Make the title specific, concise, and meaningful, to help users distinguish it from other similar topics.
- Use industry standard and commonly used words.
- Use the same terms consistently in topics that cover the same subject.
- Capitalize each significant word in the title.
- Try to limit the length to no more than eight words.

B.2 Linking

You can create hyperlinks to targets in the same DITA file, to other DITA files, or to web pages. The **@keyref** attribute provides an indirect reference to topics and other internal or external content. When the DITA content is processed, key references are resolved by using key definitions in DITA maps.

Indirect linking is handled by CACTUS through the Link > Cross-Reference function. Authors and editors are not responsible for defining keys or creating keyrefs.

B.3 Reuse

The main units of reuse in CACTUS DITA authoring are topics or entire maps.

Topics and maps may be used in more than one publication. You can also publish the same topics or map for more than one output type, such as PDF and EPUB. The content itself does not change, only the output format.

B.3.1 Maintaining the Integrity of Topics

If you mix content types in a single topic or include multiple subjects, you limit the situations in which the complete topic could be reused. All subjects in the topic must apply in any situation in which it could be reused. By separating subjects and information types, you have more flexibility to reuse information. See also Topic-Based Authoring.

B.3.2 Writing for Reuse

If explicit details are not required, leave them out. Explicit details may make it more difficult to reuse the content in situations where the details may not apply. Use encompassing phrases such as the following to indicate that details may not apply in all situations:

- such as
- including
- where applicable

Identify opportunities to create and edit content to make it reusable. For example, remove callouts on screen captures that constrain them to specific steps in a procedure so that they can be used in other procedures.

B.3.3 Reconsidering Cross-References

Limit the number of <xref> elements within your topics. Explicit cross-reference links to other topics limits the reuse of a topic to publications that also include the referenced topics.

B.4 Conditional Publishing

This is a common requirement: creating different versions of the same publication with variations based on product version or audience. The publications typically share a significant amount of common content.

DITA has powerful support for conditional publishing through the use of attributes such as **@audience**, **@platform**, and **@product**, which identify the conditional filter. The **@audience** attribute is commonly used in TD training manuals.

You can use conditional attributes to show or hide elements within topics. This allows reuse of a topic that is common between publications. An example would be using the **@audience** attribute to add a few additional statements to a common topic.

There are instances when showing or hiding an entire topic is more appropriate, however. For example, in an ITM, a topic that includes interactivity may be preferable to explain a subject, but in an FTM, the author is limited to images and text. In this case, the **@deliveryTarget** attribute could be used effectively on <topicref> elements in a map, to show or hide entire topics.

Assigning a specific value to the **@audience** conditional attribute within a topic is less flexible and less visible than assigning the **@audience** value to a topic reference in a map. In addition, by default, conditional attributes assigned to topic references in a map override conditional attributes assigned in the topic itself.

Appendix C Requesting a CACTUS Account

Access to Training Development requires a CACTUS account. To request an account, send the following information to the appropriate manager or delegate in the Training & Professional Services (T&PS) organization:

- Last name, first name
- BEMS ID
- Boeing Windows NT ID
- Organization

  Training Development (TD)
- Role required
  - Editor
  - Deployment
  - Flight Standards (FS)
  - Fleet Manager (FM) (specify major model)
  - Instructional Design (ISD)
  - Lead (specify major model)
  - SME (specify major model)
- Boeing e-mail address

The account request is then processed using the Flight Services' Application Request Management System (ARMS) and is typically created within 24 hours of being submitted to ARMS.

Appendix D Installing and Integrating oXygen

1. Open Software Express.
2. Search for **CACTUS**.
3. Double-click **CACTUS Version** [**…**].
4. Click **Start**.

   The installation begins and may take as long as 15 minutes.
5. Start oXygen.
6. If this is a new installation, skip to step 7.

   If you are updating CACTUS, you will first need to uninstall the existing CACTUS framework and plug-in:

   a. Select **Help** > **Manage add-ons**.

   b. Select all items in your add-on list.

   c. Click **Uninstall**, then **Yes**, then **OK**.

   d. Close oXygen.

   Note: You *must* close oXygen in order to complete the uninstall.

   e. Re-open oXygen.
7. Select **Help** > **Install new add-ons**.
8. For the **Show add-ons from** field, select from the drop-down list or enter

   https://cactus-app-prod.ca.boeing.com:9443/oxygen/updatesite.xml.
9. Select **CACTUS Training Development framework add on for Oxygen** and the **CACTUS plugin add on for oXygen**, and then click **Next**.
10. Review and accept the license agreement.
11. Click **Finish**, then **Continue anyway**, and then **OK**.
12. Close oXygen.

    Note: You *must* close oXygen in order to complete the installation.

    You are now ready to start using CACTUS.

Changing the CACTUS Environment

The environment that oXygen connects to is based on two factors: an environment variable and the CACTUS framework. By default, oXygen uses the CACTUS Production environment; this should not be changed unless you are a member of the testing team or have been instructed to do so by CACTUS support.

The framework is a set of CACTUS-specific functions that oXygen uses. A different version of the framework exists for each environment. Using the wrong environment can result in data corruption or undesired functionality.

Changing the CACTUS environment that oXygen uses consists of three main steps:

1. Changing the environment variable.
2. Adding the update site URL (not required if you have previously added the environment).
3. Installing the framework.

You must have Administrator rights on your computer to make these changes. If you do not have Administrator rights, you can request temporary Administrator rights at http://upep.web.boeing.com/.

1. Close all Firefox windows, if any are open.
2. On the **Start** menu, right-click **Computer** and choose **Properties** on the pop-up menu.
3. In the **Control Panel**, select **Advanced system settings**.
4. In the **System Properties** dialog box, click **Environment Variables**.
5. Look for a user variable named **com.boeing.cactus.environment**.
    - If the variable exists, complete the following:
        a. Select the variable
        b. Click **Edit**.
        c. Change the variable value to the environment that you want to use (e.g., for the Pre environment, enter **pre**, for the Test environment, enter **test**). If you are changing back to the production environment, enter **prod**.
        d. Click **OK**.

- If the variable does not exist, create a new system variable:

    a. Click **New** under the user variable section.

    b. In the **Name** field, type **com.boeing.cactus.environment**.

    c. For the variable value enter the environment you want to use (**prod**, **pre**, **test**, or **dev**).

    d. Click **OK**.

    e. Look for a user variable named **com.boeing.cactus.kerberos**.

    f. Repeat the steps for creating a new variable, setting the value for **com.boeing.cactus.kerberos** to **true**.

6. Click **OK** twice to close the dialog boxes.

7. Close the **Control Panel**.

Appendix E Installing Firefox and Java

Both Firefox and Java are available from Software Express:

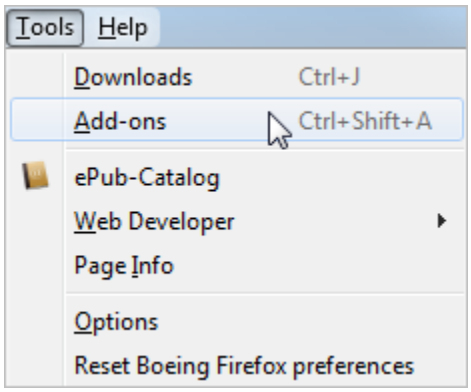
- Firefox – go to **Common Authorized Business Software** > **A-F Catalog** > **Firefox 52.3.0 – Mozilla**.
- Java – go to **Common Authorized Business Software** > **G-L Catalog** > **Java** > **Java Rule Set 1.0.14**.

If a Java error is displayed when you open CACTUS, you may need to update your version of Java:

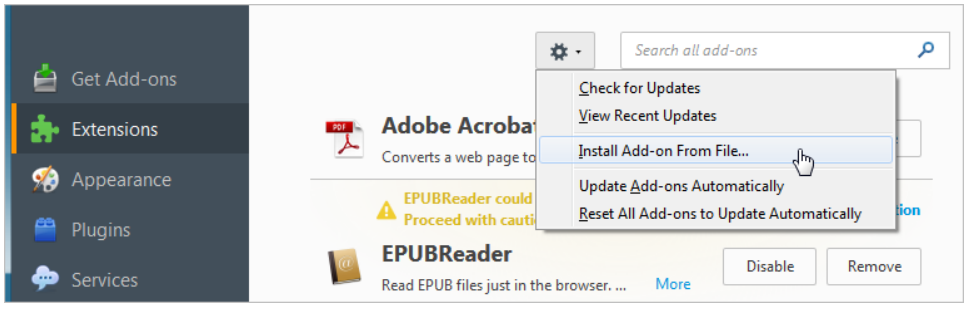
1. Download Prod CACTUS Firefox Add-on api from this URL:

   https://cactus-app-prod.ca.boeing.com:9443/browser-extension/cactus-client-extension.xpi.

2. Open Firefox and go to **Tools** > **Add-ons**.

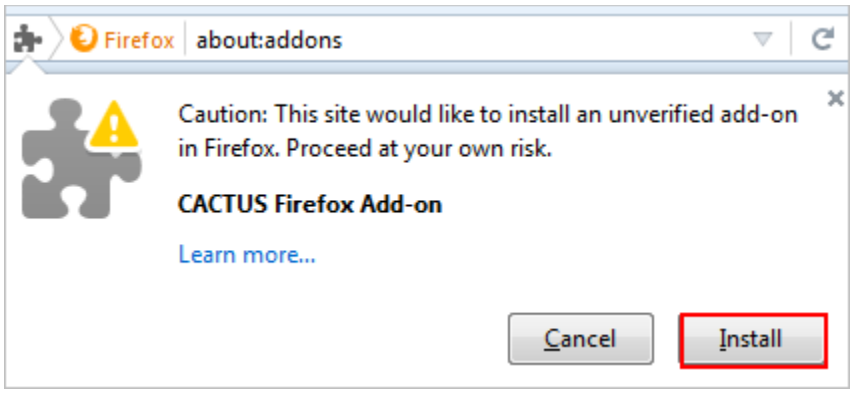


3. Click **Extensions**.
4. Select **Install Add-on From File**, and then select the downloaded api file.



5. Select **Install**.

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6. Restart Firefox after installation.

   CACTUS Firefox Add-on will be listed under **Extensions**.



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Appendix F Contacts

- To request a CACTUS account, contact the appropriate manager or delegate in the Training and Professional Services (T&PS) organization. See Requesting a CACTUS Account appendix.

- For application support, e-mail the CACTUS Help Desk: GRPCACTUSHelpDesk@exchange.boeing.com.

Appendix G Resources

The following resources provide fundamental “getting started” information about DITA.

- Introduction to the Darwin Information Typing Architecture. An early (2002) overview by Don Day, Michael Priestely, and David Schell, the architects of DITA at IBM.

- Introduction to DITA: Second Edition. Comprehensive introduction by JoAnn Hackos. Expanded revision of the original edition to cover DITA 1.2.

- DITA for Practitioners Volume 1: Architecture and Technology. Overview of DITA processing and architecture for beginning and experienced practitioners, by DITA expert Eliot Kimber. More advanced than *Introduction to DITA: Second Edition*.

- DITA Open Toolkit User Guide. User documentation for the DITA OT, version 2.2.

Revision Record

Release/Revision	**NEW**
Contract Number (if required)	
Limitations	
Description of Change	
Authorization for Release	

AUTHOR:	Richard Sanford	NA
	Name	BEMSID
	GG-GG-82E4	03/15/2017
	Organization Number	Date
APPROVER:	Kim Toy	2896565
	Name	BEMSID
	66-CM-Z887	03/15/2017
	Organization Number	Date
DOCUMENT RELEASE:	NA	
	Name	BEMSID
	Organization Number	Date

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Release/Revision	**A**
Contract Number (if required)	
Limitations	
Description of Change	Added sensitivity marking. Added more details to features. Updated screenshots.
Authorization for Release	

AUTHOR:	Pamela Widlitz	271475
	Name	BEMSID
	GG-GG-82E4	10/17/2018
	Organization Number	Date
APPROVER:	Kim Toy	2896565
	Name	BEMSID
	66-CM-Z887	10/17/2017
	Organization Number	Date
DOCUMENT RELEASE:	NA	
	Name	BEMSID
	Organization Number	Date

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